# Waiting for Closure: Deep Networks Still Do Not Complete Contours

Benjamin TenWolde
University of California San Diego
bendotten@gmail.com

**Abstract**

Whether deep networks exhibit the Gestalt law of closure has been answered both ways, using stimuli that are rarely sized to the network's sampling grid and measures that are rarely checked against observers that cannot close contours. We built a patch-aware test battery in which every gap, inducer and dash is specified in units of the model's patch size and rendered natively, with matched controls that rotate, scramble or re-orient the same pieces. We trained and tested 24 networks: frozen and fine-tuned CNNs and vision transformers, looped (weight-tied, recurrent-depth) transformers, and fGRU-H, $\gamma$-net and deep fGRU-H/fGRU circuits built from the published fGRU cell, including stacks of up to eight recurrent layers behind a CNN stem. Illusory contours are at best weakly represented: ViT-B/16 variants show a small but consistent advantage of Kanizsa figures over rotated-inducer controls (+0.08 to +0.14 across three fine-tuned seeds, +0.10 frozen), with Kanizsa accuracy of at most 0.39 (chance 0.10) against 1.00 on complete outlines; DINOv2-S shows an inconsistent advantage (+0.01 to +0.15 across seeds), and in the other networks it stays at or below +0.07, far from the fragment effects. Fine-tuned transformers do integrate fragmented outlines, and the arrangement carries the effect: scrambling piece positions while keeping orientations lowers accuracy by +0.46 to +0.65, and pieces placed correctly but misoriented are not recognized. Neither looping nor recurrence consistently adds closure: in five of six recurrent circuits a control trained with a single iteration matches or exceeds the eight-iteration network, although the recurrent states in the gap become more similar to those for the complete outline over iterations. Finally, we audited published closure measures with null observers. A blur reproduces the illusory-brightness signature reported for a predictive-coding network; a pixel-growth observer, which only sees how far contour pixels lie from their centroid, scores +0.289 on the closure measure of Kim et al. [22], several times any CNN, and rotating each corner in place about the centroid of its own pixels, which removes the radial difference this observer detects, reduces the measure to +0.005 and +0.005 in the two networks previously reported to show closure most consistently. The configural-effect (CE) measure survives null observers and is specific to closing arrangements, but it generally grows as the outline approaches completion, its advantage over a non-closing arrangement gains nothing further, on average, from a fully closed square, and changes with inducer shape and size at a fixed ratio of luminance-specified to total edge length, which predicts the clarity of human illusory boundaries across display sizes [45]. In 9 of 10 CNN and recurrent networks the mid-gap closure index halves at gaps of 6–50 px, inside the network's measured effective receptive field. Networks trained specifically to trace contours, including weight-tied looped transformers, link aligned pieces only across gaps a few pixels wider than in training and label almost none of the missing contour as contour. In the transformers whose classifications favour Kanizsa figures over rotated inducers, activation patching locates that evidence in the inducers' tokens, not in the tokens of the gaps. Recurrent circuits trained to spread a grouping label from a cue across objects in natural images, reimplemented after Mollard et al. [33], label the surface of a Kanizsa figure only when the gaps between inducers are a few pixels wide; single-step controls do as much, and the limit does not move with the circuits' kernel size. Deep networks group nearby,

consistently oriented contour pieces; they do not close gaps, and illusory contours are at most weakly represented.

## 1 Introduction

Closure, the tendency to perceive an incomplete figure as a whole, is among the oldest Gestalt laws [21, 50] and a standard test of whether artificial vision organizes the world as people do [22, 54]. Studies of deep networks disagree. Similarity-based analyses report closure in ImageNet-trained CNNs [22, 54]; accuracy-based protocols report its absence [53]; classification images suggest networks rely on inducer orientation rather than illusory edges [4]; and recurrent predictive-coding networks have been reported to perceive Kanizsa squares [35]. Classifiers of regular polygons degraded at their corners or at the middle of each edge behave unlike human observers, who recover shapes with deleted midsegments but not with deleted vertices [6]: ResNets suffer more from edge than from corner degradation, and a ViT equally from both [48]. For vision-language models, whose image encoders are vision transformers, answers to classic illusions persist when the illusion-inducing elements are inverted, which Sun et al. [46] trace to different causes in different models, from recall of the canonical answer to limits of visual processing.

Two problems run through this literature. First, stimuli are drawn in pixels at arbitrary scales and then resized, so the size of a gap relative to a vision transformer's patches or a CNN's receptive field is uncontrolled. A six-pixel gap need not leave any 16-pixel patch empty, although it remains detectable within its patch; a forty-pixel gap may exceed every receptive field below the last stage. Second, measures are checked against some null controls, such as untrained networks [22], but not against simple observers that *cannot* close contours yet share the stimuli's low-level statistics, such as the radial distribution of contour pixels. If a blur, a filter bank or an untrained network produces the same signature, the signature is not evidence of closure.

We address both problems. Our contributions are: (i) a patch-aware battery with geometry in patch units and matched controls (rotated, position-scrambled and orientation-swapped pieces, Kanizsa inducers, and dashed, noisy linking segments); (ii) a comparison of 24 networks spanning feed-forward, looped-transformer and horizontal/top-down recurrent architectures built from the published fGRU cell, each recurrent circuit but one with a matched single-iteration control; (iii) a null-observer audit of four published closure measures; and (iv) a faithfulness check of the saliency overlays commonly used to argue that networks "see" a shape.

## 2 Stimuli

Networks classify regular polygons with 3 to 12 sides, the ten-way task of Zhang et al. [53]. All lengths are set in units of the patch size $P$ (16 px for ViT-B/16 and the CNNs, 14 px for DINOv2) and rendered natively at 224 px with 4× supersampling; images are never resized. The circumradius is $4.5P$ for every class, so overall size carries no class information. Training uses complete outlines with random rotation, scale (±15%), position ($\pm P$) and background polarity. The test battery (Fig. 1; about 38,000 images per patch size) contains: complete outlines; *mid-gap fragments*, with a gap of $g \in \{0.25, \ldots, 4\}P$ removed from the middle of each edge (the manipulation of Zhang et al. [53], but in absolute patch units; midsegment deletion in 6; edge degradation in 48); *corner-removed fragments*, in which contour within $g/2$ of every vertex is removed so that corner angles cannot be counted (vertex deletion in 6; corner degradation in 48); *Kanizsa figures* with inducer radius $r \in \{0.5, \ldots, 2\}P$; and *dashed links*, in which the Kanizsa gap contains a real dashed segment (spacing 0–$2P$, contrast 0–1, pixel noise only on the segment). Every fragment condition has a

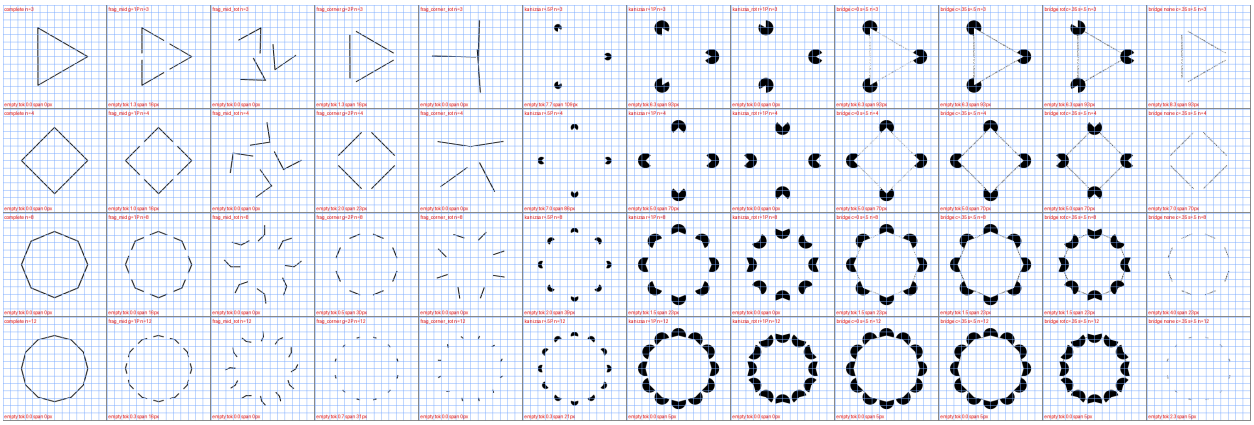


**Figure 1:** Patch-aware stimuli at 224 px, shown with the 16 px patch grid. Columns: complete outline; gaps in the middle of every edge and their rotated control; corners removed and their rotated control; Kanizsa figures with inducer radius 0.5 and 1 patch and the rotated-inducer control; dashed linking segments in the Kanizsa gap with inward, rotated and no inducers. Rows: 3, 4, 8 and 12 sides. Red text gives the number of empty patches each gap crosses.

matched control, a disordered control in the sense of Kim et al. [22], in which each piece is rotated about a point inside it (a straight piece about its midpoint; a corner piece of a mid-gap fragment about the centroid of the triangle formed by its vertex and its two ends) by a fixed pseudo-random angle (the rotated control). *Contour pixels* are the pixels of the physically specified contour [45]; a patch is *empty* if it contains none. Each image is shifted by 0, $\frac{1}{4}$, $\frac{1}{2}$ or $\frac{3}{4}$ of a patch relative to the patch grid, and its geometry, gap masks and the number of empty patches each gap crosses are recorded.

Two further test-only controls separate closure from orientation statistics (Fig. 2B,C): *scrambled* stimuli move each piece to a random position inside the shape's disk while keeping its orientation and length; *cue conflict* stimuli, a position–orientation analogue of the shape–texture cue conflict of Geirhos et al. [15], keep each corner-removed piece's midpoint but give it the edge orientation of the polygon with one more side, so positions imply $n$ sides and the orientations are $n$ of the $n{+}1$ edge orientations of the polygon with one more side (for 12-sided polygons, the 11-sided one).

## 3 Models and protocol

**Frozen backbones** follow the protocol of Zhang et al. [53], who replace the last layer of a frozen network by a fully connected layer (Adam, $10^{-3}$, 25 epochs): VGG16, ResNet50, and, not tested by them, ViT-B/16 and DINOv2-S/14. **Fine-tuned** ResNet50, ViT-B/16 and DINOv2-S/14 are trained end to end. **Looped transformers** split a pretrained ViT into prelude (blocks 0–3), a weight-tied core (blocks 4–7) applied $T$ times, and coda (blocks 8–11), re-injecting the prelude output through a zero-initialized linear map so that $T{=}1$ is exactly the pretrained model (prelude, core and coda are the terms of 14; iterating a weight-tied block follows 11, 16); $T$ is sampled from $\{1, \ldots, 4\}$ during training and swept to 12 at test. **Horizontal recurrence** uses the fGRU cell [28, 30] as specified in the appendix of Linsley et al. [30]: attention-modulated suppression gate, a gate that updates the recurrent state, channel-symmetric horizontal kernels, instance normalization of the gate inputs and the published initialization. Our adaptations are a VGG16 front end (conv2 features, stride 4), a convolutional readout (three stride-2 $3 \times 3$ convolutions with ReLUs, global average pooling and a linear layer; a $1 \times 1$ convolution in the deep hierarchies and stacks below),

unrolling for $T$=8, and, for the $\gamma$-net, two levels with max-pooled bottom-up connections. The models are a horizontal-only fGRU circuit (fGRU-H; the hGRU of Linsley et al. [28] is a different cell), an fGRU-H with a $1 \times 1$ readout that adds no spatial convolution after the recurrence, and a two-level $\gamma$-net whose top-down fGRUs update the lower layer from a projection of the upper one, read out from the lowest layer. **Deep fGRU-H/fGRU** hierarchies place fGRU layers at the four VGG stages: an eight-layer fGRU-H (two per stage, bottom-up only) and a four-layer fGRU with top-down connections, unrolled for $T$=8. **Divisive-normalization stacks** keep a CNN only at the input (VGG16 conv1–2, stride 4) and then stack 4 or 8 fGRU layers at that single resolution, each preceded by divisive normalization [18] with learned channel weights, as in generalized divisive normalization [5], over a fixed $5 \times 5$ average pool of our choosing; layers are linked by $1 \times 1$ convolutions, so spatial integration beyond the stem comes from the recurrent horizontal kernels, the $3 \times 3$ attention kernel of the suppression gate and the divisive-normalization pool, and top-down fGRUs update each layer through a zero-initialized gate. Every recurrent model except the fGRU-H with a $1 \times 1$ readout has a single-iteration control (trained with $T$=1). All networks except the frozen ones are trained for 15–20 epochs on 6,000 complete polygons and reach 1.00 validation accuracy; frozen backbones reach 0.97–1.00. Transformer inputs are normalized with the statistics of each checkpoint's own pretraining.

**Measures.** The *closure index* is accuracy on a stimulus minus accuracy on its rotated control. We also report the closure measure $\bar{C}$ of Kim et al. [22] on our stimuli (cosine similarity to the matched complete image, stimulus minus control), the linear probe of Zhang et al. [53] trained at every layer [a linear classifier probe; 3], the effective receptive field [31], and, for recurrent models, the *gap similarity*, $\bar{C}$ restricted to the gap region of the recurrent state (for Fig. 9B: one-patch corner-removed fragments, all cells the missing contour crosses, the first recurrent layer; the gap analysis uses mid-gap fragments, cells empty in both images and the layer the classifier reads): $\cos(h_t^{\text{stim}}, h_t^{\text{complete}}) - \cos(h_t^{\text{ctrl}}, h_t^{\text{complete}})$ at each iteration $t$.

# 4 Results

## 4.1 Illusory contours are at most weakly represented

Across all networks that learned the training task (first training seed of each), accuracy on Kanizsa figures lies between 0.04 and 0.39 (chance 0.10), and the Kanizsa closure index reaches at most +0.10 (ViT-B/16 (frozen); Fig. 2A). Turning every inducer in place by 15° halves the transformers' Kanizsa advantage, but part of it remains at 45°, so it does not require the inducer edges to line up (Appendix C). Across three to five training seeds of three networks (Table 4), fragment closure indices and arrangement effects (accuracy in place minus accuracy with scrambled positions; see below) replicate (largest s.d. 0.03 for corner-removed fragments and 0.05 for the arrangement effect); the Kanizsa index is small but consistent in fine-tuned ViT-B/16 (+0.08 to +0.14, Kanizsa accuracy 0.18–0.35), inconsistent in DINOv2-S (+0.01 to +0.15) and near zero in ResNet50 (-0.05 to +0.07), always against 1.00 accuracy on complete outlines (Table 4). The frozen and looped ViT-B/16 show indices of the same size (Table 2), so ViT-B/16 carries a weak illusory-contour signal that is consistent across seeds only in this network. Class-specific relevance maps [9] for the top-1 class illustrate this: in Fig. 7 Kanizsa triangles and squares are labelled as hexagons, and their rotated inducers receive the same label. Dashed linking segments raise accuracy about as much with aligned inducers as with no inducers: from contrast 0 to 1 the median gain over 22 networks is +0.14 with aligned inducers and +0.11 without inducers (difference -0.01; above +0.05 in 6 networks and below −0.05 in 8), whereas rotated inducers suppress it (+0.01). Noise was drawn independently for each image.

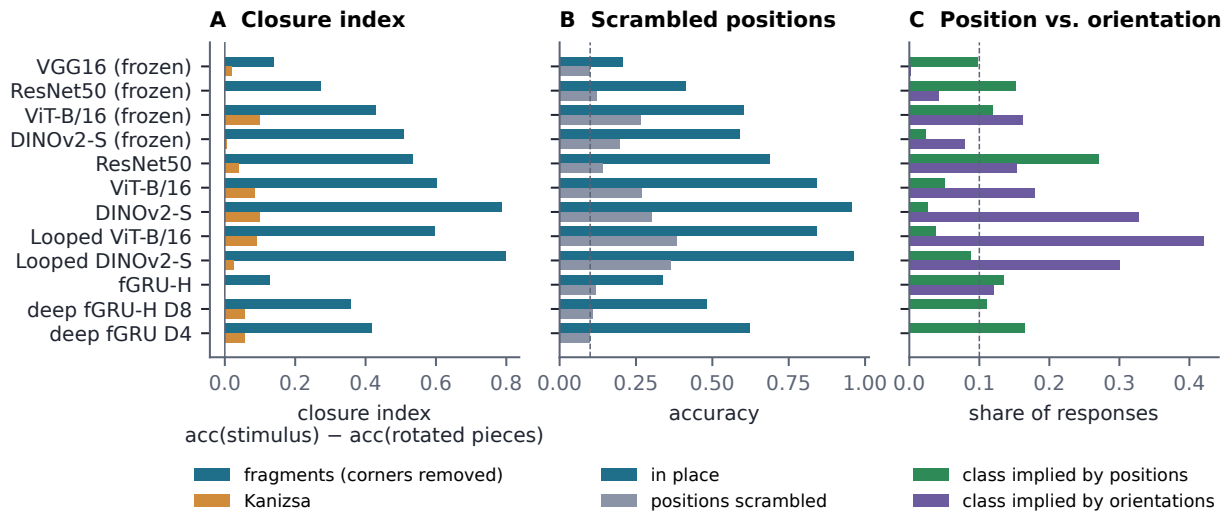



**Figure 2:** (A) Closure index for corner-removed fragments (all gap sizes) and Kanizsa figures. (B) Corner-removed fragments in place versus the same pieces with scrambled positions and unchanged orientations; dashed line, chance. (C) Cue-conflict stimuli: how often the network picks the class implied by the pieces' positions versus their orientations.

### 4.2 Fine-tuned transformers integrate fragments, and the arrangement matters

With corners removed, fine-tuned ViT-B and DINOv2 (plain and looped) classify fragmented outlines at 0.84–0.96 in the orientation-control battery (a separate test set of corner-removed fragments in place, scrambled and cue conflict, gaps of 0.5–2$P$), and their closure index over all gap sizes of the main battery is +0.60 to +0.80. Frozen backbones reach half to 70% of that, and fine-tuning raises the index 1.4–2× within each family. Scrambling the pieces' positions while keeping their orientations drops the fine-tuned transformers to 0.27–0.39 (Fig. 2B): the arrangement adds +0.46 to +0.65 beyond orientation statistics. When positions and orientations conflict, these networks pick the class implied by positions only 0.03–0.09 of the time and the class implied by orientations 0.18–0.42 (Fig. 2C). Pieces must therefore be both placed and oriented along the contour, the signature of contour integration by good continuation [13] rather than of position or orientation alone.

The manipulation used by Zhang et al. [53], which keeps the corners, behaves differently: scrambling corner pieces costs fine-tuned transformers only +0.12 to +0.17, so that test is largely solved from corner angles and cannot separate closure from corner reading.

### 4.3 Patch geometry decides who can bridge a gap

Plotted against the number of empty patches a gap crosses, accuracy separates the network families (Fig. 3); we did not test whether this count predicts accuracy better than gap size. Averaging, for 3–6-sided polygons, over gaps that cross three or more empty patches (bins of 3, 4, 5 and 6 or more empty patches weighted equally), fine-tuned transformers score 0.78–0.99, while the CNNs and the representative fGRU-H/fGRU circuits score 0.10–0.39.

**The bridgeable gap is far smaller than the receptive field.** For each network we define a critical gap $g^*$, the gap at which the mid-gap closure index falls to half its value at the smallest gap, and compare it with the measured $\mathrm{ERF}_{50}$ for CNNs and recurrent circuits (our summary of the effective receptive field of Luo et al. [31]: the radius around the gradient centroid that contains half

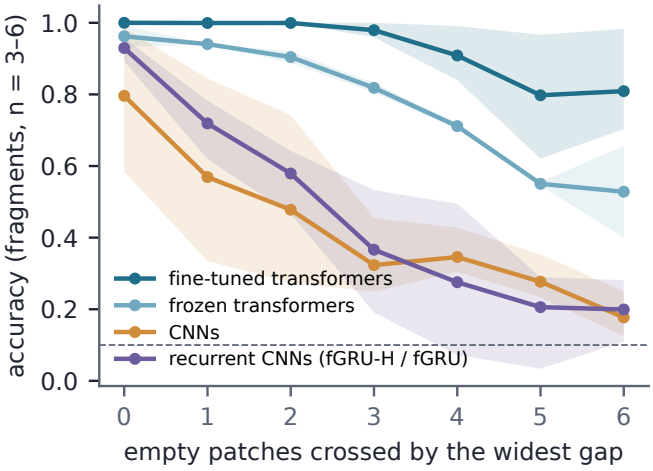



**Figure 3:** Accuracy on fragments ($n = 3$–6, all gap sizes valid) against the largest number of empty patches crossed by any gap. Lines: group means; bands: range across networks.

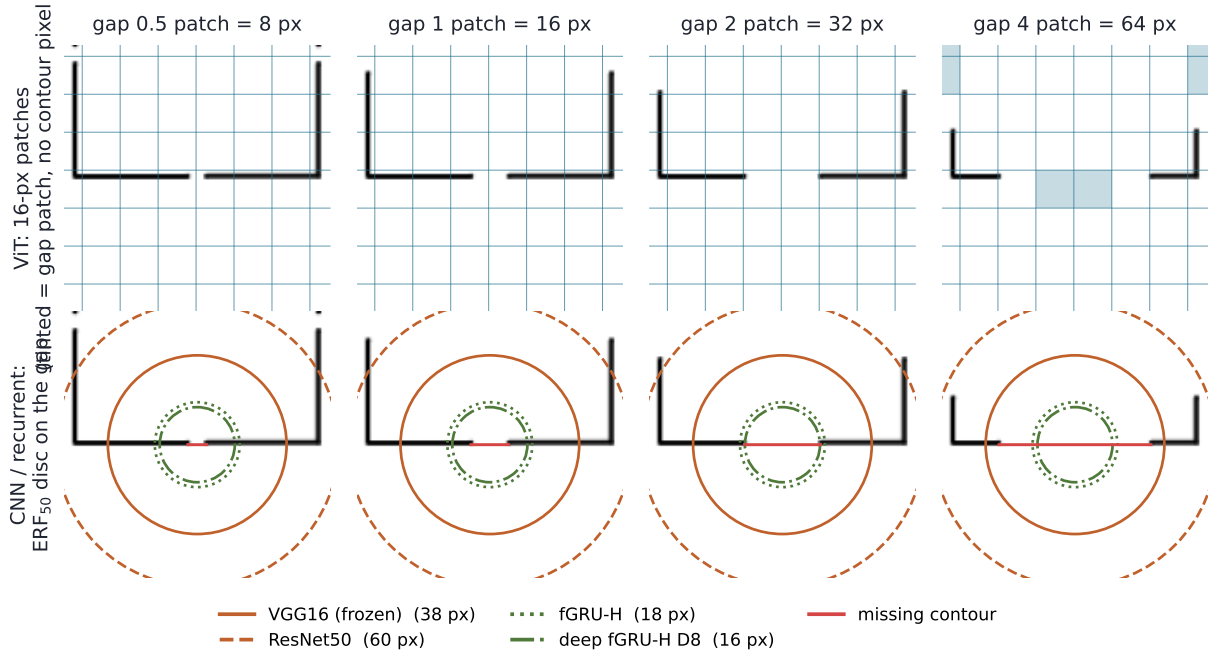



**Figure 4:** The same mid-gap fragment at four gap sizes, zoomed on one gap. Top: the 16 px patch grid a ViT-B/16 tokenizes; tinted patches are crossed by the missing contour and contain no contour pixels. Bottom: the measured 50% effective receptive field ($ERF_{50}$) of four networks, centred on the gap (red: missing contour). Even the 64 px gap lies inside the $ERF_{50}$ of two of the four networks shown.

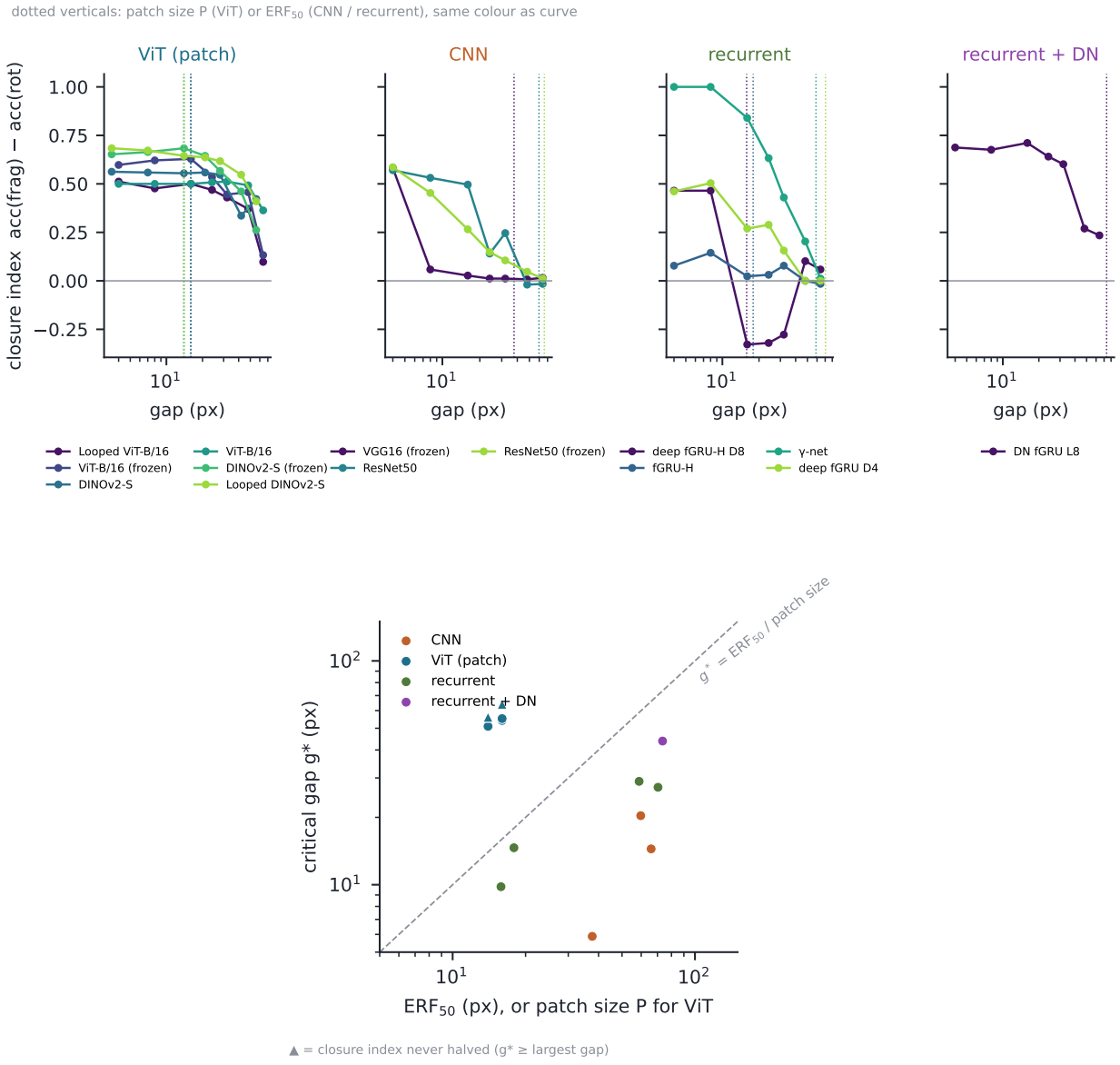



**Figure 5:** Top: mid-gap closure index against gap size, by family (one network per recurrent family; all networks in Table 3); dotted verticals mark each network's $\mathrm{ERF}_{50}$ (patch size for transformers). Bottom: critical gap $g^*$ (closure index halved) against $\mathrm{ERF}_{50}$ or patch size; dashed line, equality. Excluding single-iteration controls, all CNN and recurrent networks but one (the fGRU-H with a $1 \times 1$ readout) lie below the diagonal; including them, 14 of 16 do.

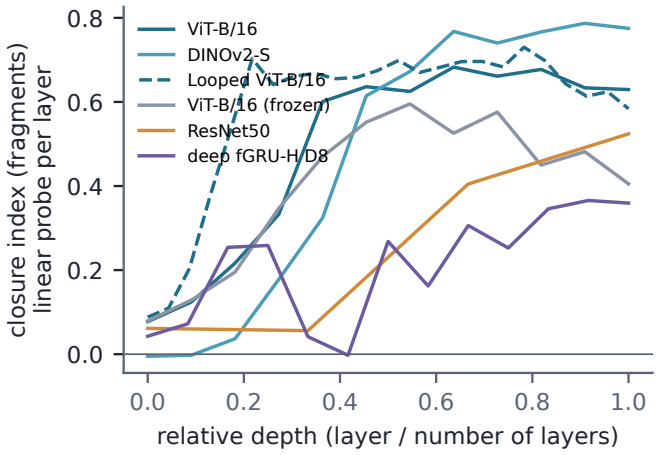



**Figure 6:** Fragment closure index from a linear probe trained on complete outlines at every layer, against relative depth. For the looped ViT, depth counts executed blocks (loops unrolled).

of the absolute gradient of a central unit with respect to white-noise input) and with the patch size for transformers (Figs. 4, 5). A transformer's attention can span the whole image, and the analogue of receptive-field size proposed by Dosovitskiy et al. [12], the mean attention distance, varies across heads and layers; we therefore use the patch size, the resolution at which the gap is tokenized, as the reference. We compute the index on the fixed set of 3- to 6-sided polygons, for which every gap size is valid; pooling all classes would change the mix of classes with gap size, because larger gaps exceed the edges of many-sided polygons. For CNN and recurrent networks other than the single-iteration controls, $g^*$ lies between 6 and 50 px; it marks where the index halves, and a smaller residual effect can persist beyond it. For 9 of 10 CNN and recurrent networks it is smaller than $\mathrm{ERF}_{50}$ (median ratio 0.44; 14 of 16 including the single-iteration controls), and its rank correlation with $\mathrm{ERF}_{50}$ is only +0.33; the exception, excluding single-iteration controls, is the fGRU-H with a $1 \times 1$ readout, whose readout adds no spatial convolution after the recurrence and which has the largest fragment closure index of any recurrent circuit (+0.63). In 9 of 10 CNN and recurrent networks (excluding single-iteration controls) the index therefore halves at gaps whose full length fits inside the measured $\mathrm{ERF}_{50}$, a radius; a smaller effect can remain at larger gaps. $\mathrm{ERF}_{50}$ is measured on noise images, and for the deep hierarchies at the first recurrent layer, so this comparison is descriptive: it shows that the gap lies within the region that influences a unit, not that the network uses that region for the task. Transformers behave differently: 3 of 6 keep more than half their mid-gap index up to the largest gap tested (four patches), and the others halve at 3.4–3.6 patches. Mid-gap fragments keep the corners, which these networks read directly, so this index overstates how far they bridge gaps.

### 4.4 Where closure emerges, and what the maps show

Applying the linear probe at every layer (Fig. 6), fragment closure in fine-tuned transformers rises through the middle blocks and peaks late (ViT-B +0.68, DINOv2-S +0.79). In the looped ViT it reaches +0.73 by the fourth pass through the shared core and gains at most about 0.02 with further iterations, so additional loops add depth without adding closure. Frozen ViT-B peaks at +0.60 and fine-tuned ResNet50 at +0.52 in its last stage. Relevance maps make the dissociation between fragments and illusory contours visible (Fig. 7): for fragmented outlines, relevance follows the whole contour including its gaps, whereas for Kanizsa figures it stays on the inducers, and Kanizsa triangles and squares are labelled "hexagon" whether their inducers face inward or are

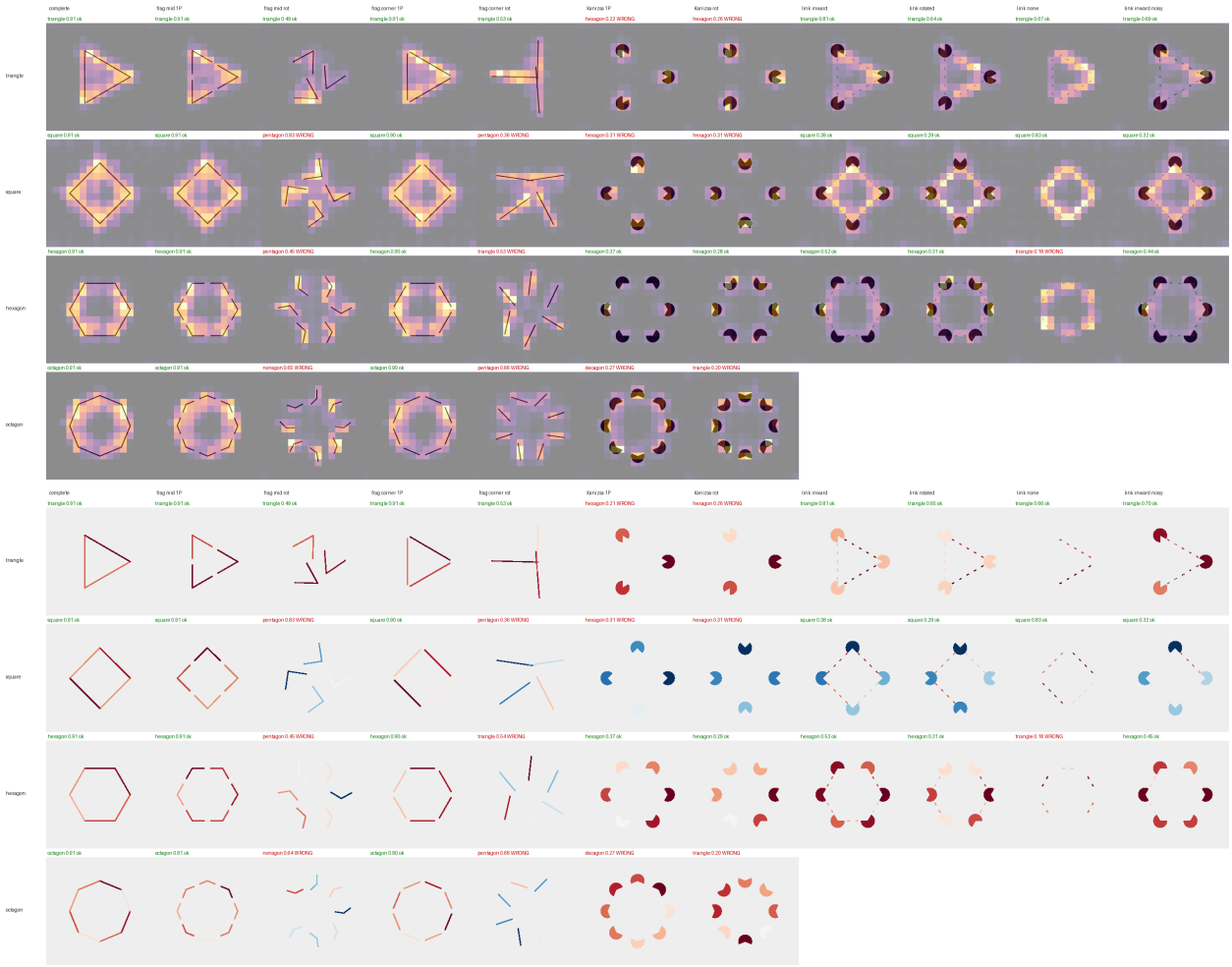


**Figure 7:** Fine-tuned DINOv2-S. Top: class-specific relevance maps [9] for the top-1 class per condition (columns) and true class (rows); cell titles give the prediction and its probability. Bottom: Shapley values [44] of the stimulus's pieces for the true label (red supports, blue opposes). Illustrative only: these maps are weakly faithful (Section 4.11); the behavioural evidence for integration is in Fig. 2.

rotated.

### 4.5 Nothing is represented between the inducers

The published studies probe representations; the most direct probe is the activity in the empty cells where an illusory edge would be. For 20 networks (all with spatial feature maps) we read every layer, block or iteration in two ways, both fitted only to complete outlines and never to Kanizsa figures: a linear *contour probe*, a logistic regression that labels feature-map cells as contour or empty (trained on two of the four grid shifts and tested on the others), and the raw activity of the 5% most *contour-selective units*, those that best separate contour cells from empty cells, scaled so that an empty cell reads 0 and a drawn contour 1. We compare the illusory edge with physically drawn versions of the same shape: a real line of contrast 3–100% drawn between the same inducers, dashes, a faint filled surface, some sides drawn and the others left empty, and a side that fades to nothing by its middle (Fig. 8A); every display also appears with the inducers rotated in place. A read-out is used only where it rises with the contrast of the real line (rank correlation $\geq 0.8$), which holds for

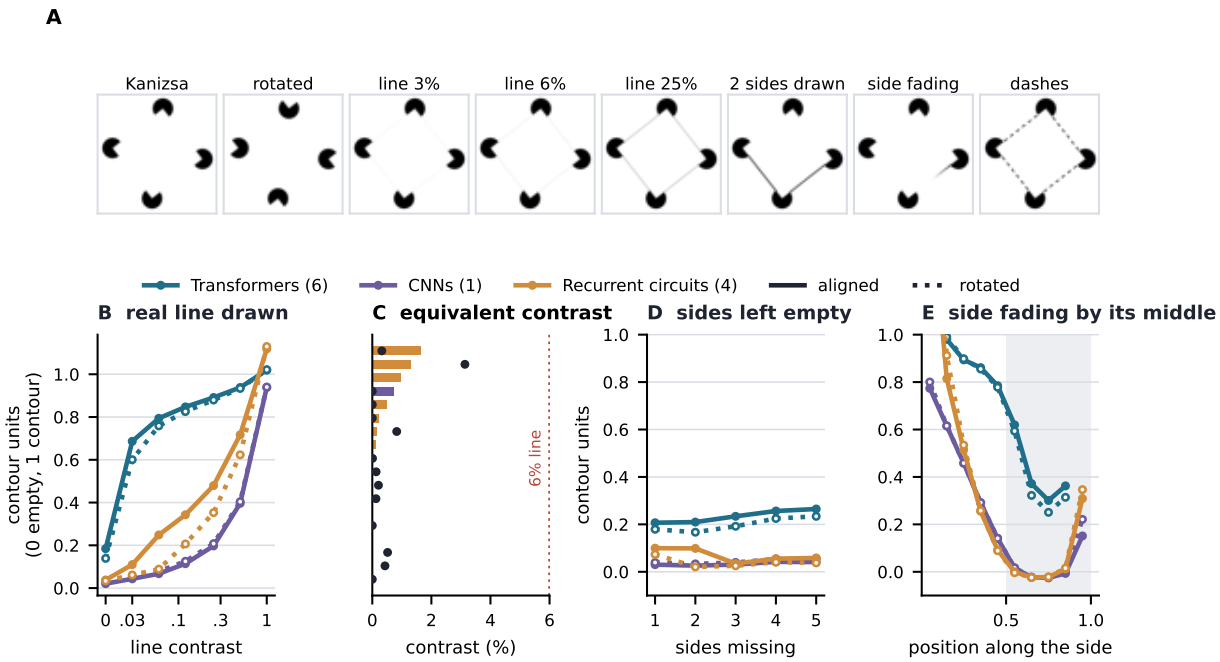



**Figure 8:** The illusory edge against physically drawn versions of the same shape. (A) Displays: the Kanizsa square and its rotated control, a real line of increasing contrast between the same inducers, some sides drawn and the others left empty, a side fading to nothing by its middle, dashes. (B) Contour units (the 5% most contour-selective units of a layer, scaled so that an empty cell is 0 and a drawn contour 1) in the empty cells on the illusory edges, against the contrast of a real line drawn there; contrast 0 is the plain Kanizsa figure (solid) or its rotated control (dotted). Bold: median of each family; networks whose read-out does not rise with line contrast are omitted. (C) Equivalent contrast of the illusory edge per network: bars, contour probe; dots, contour units. (D) Contour units on the sides left empty, against how many sides are missing (the others drawn). (E) Contour units along a side drawn at full contrast at one end and fading to nothing by its middle; grey, no contour pixels.

the probe in 18 networks and for the units in 14. The *equivalent contrast* of the illusory edge is the contrast of a real line that adds as much response as aligned inducers add over rotated ones.

A real line at 6% contrast raises the contour units by a median +0.44; aligned inducers raise them by +0.02 over rotated ones (at most +0.07; Fig. 8B). The illusory edge is therefore worth at most 1.7% of a real line's contrast to the contour probe and at most 3.1% to the contour units (Fig. 8C). Sides left empty stay empty however many of the others are drawn (aligned minus rotated, +0.02; Fig. 8D), and along a side that fades out the response ends where the contour pixels end: in the empty half it is 0.18 with aligned and 0.18 with rotated inducers (Fig. 8E). In real gaps the same probe responds: in the empty cells of one-patch gaps between fragments, aligned pieces add a median 0.43 of the drawn-contour response over rotated pieces, against 0.02 between inducers at inducer radius 1 patch (largest, DINOv2-S, 0.34, from cells next to the inducers). One patch inside the virtual shape, cells resemble the complete shape slightly more with aligned than with rotated inducers (+0.09; one patch outside, +0.02), but these cells lie next to the inducers' inward-facing notches, so the difference does not separate a filled surface from the notches' own edges. Shapley interactions among the inducers, which unlike Shapley values of image regions can register completion, give the same answer: relative to a 180° control, fragment corners are superadditive but Kanizsa inducers are not, except weakly in transformers (Appendix D).

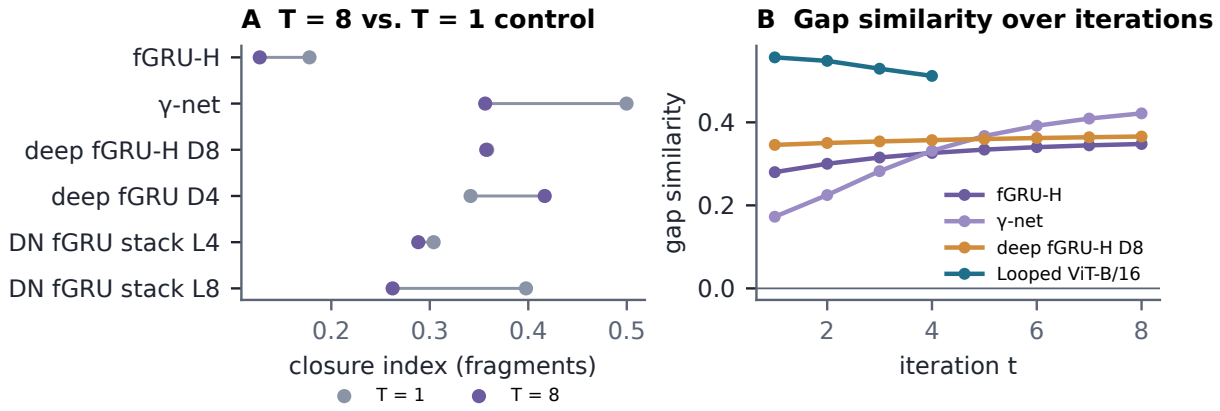



**Figure 9:** (A) Fragment closure index for the six recurrent circuits that have single-iteration controls (T = 8) and those controls (T = 1). (B) Gap similarity in the gap region of the recurrent state (first recurrent layer, one-patch corner-removed fragments) across iterations.

### 4.6 Recurrence does not consistently add closure

Looping a fine-tuned transformer changes the fragment closure index by -0.01 to +0.01. **Recirculation.** Recirculation [34] turns a frozen transformer into a dynamical system by leaking a renormalized deep-layer activation back into a shallow layer at every step. On a single static image there is no step dimension, so what we test is training-free recurrence in depth (looping, in the terms of 34) with recirculation's mixing rule, in which the source activation is rescaled to the destination's norm before being added: fine-tuned and frozen ViT-B/16 and DINOv2-S re-process the same image with the previous pass's deep state mixed into the shallow layer, over 60 settings per network of source layer, destination layer, mixture coefficient and number of passes. Among settings that lowered accuracy on complete outlines by at most five percentage points, none raised the Kanizsa closure index by more than +0.03 or the fragment index by more than +0.03 (best Kanizsa index of a fine-tuned network +0.14), small differences, selected post hoc, whose reliability we did not estimate. For fGRU-H and fGRU circuits, recurrence increases the gap similarity: it rises from 0.28 to 0.35 (fGRU-H), 0.17 to 0.42 ($\gamma$-net) and 0.35 to 0.37 (eight-layer deep fGRU-H) over iterations (Fig. 9B). Yet increased gap similarity does not consistently accompany a higher closure index. Recurrence does not consistently improve the fragment closure index: in 5 of 6 circuits the control trained with a single iteration is within 0.02 of, or above, its eight-iteration counterpart (for example the fGRU-H, +0.13 versus +0.18; Fig. 9A), and the one gain, +0.08 in the deep fGRU D4, is from a single seed. **Divisive normalization with a CNN stem only.** When the CNN is reduced to a stem and every further layer is a recurrent fGRU preceded by divisive normalization, the corner-removed closure index is +0.29 for the four-layer and +0.26 for the eight-layer stack, well below the fine-tuned transformers; the single-iteration controls score +0.30 and +0.40, and the Kanizsa index of all 4 stacks stays at or below +0.01. Across recurrent circuits the Kanizsa index is at most +0.06 (single-iteration controls +0.04), so recurrence adds no illusory-contour signal either.

### 4.7 Networks trained to trace contours link pieces but do not complete them

Our recurrent circuits were trained on complete outlines, a task that never demands long-range integration. We therefore probed networks trained for exactly that: hGRU, xhGRU (an hGRU whose per-iteration batch normalization is replaced by group normalization [51] shared across

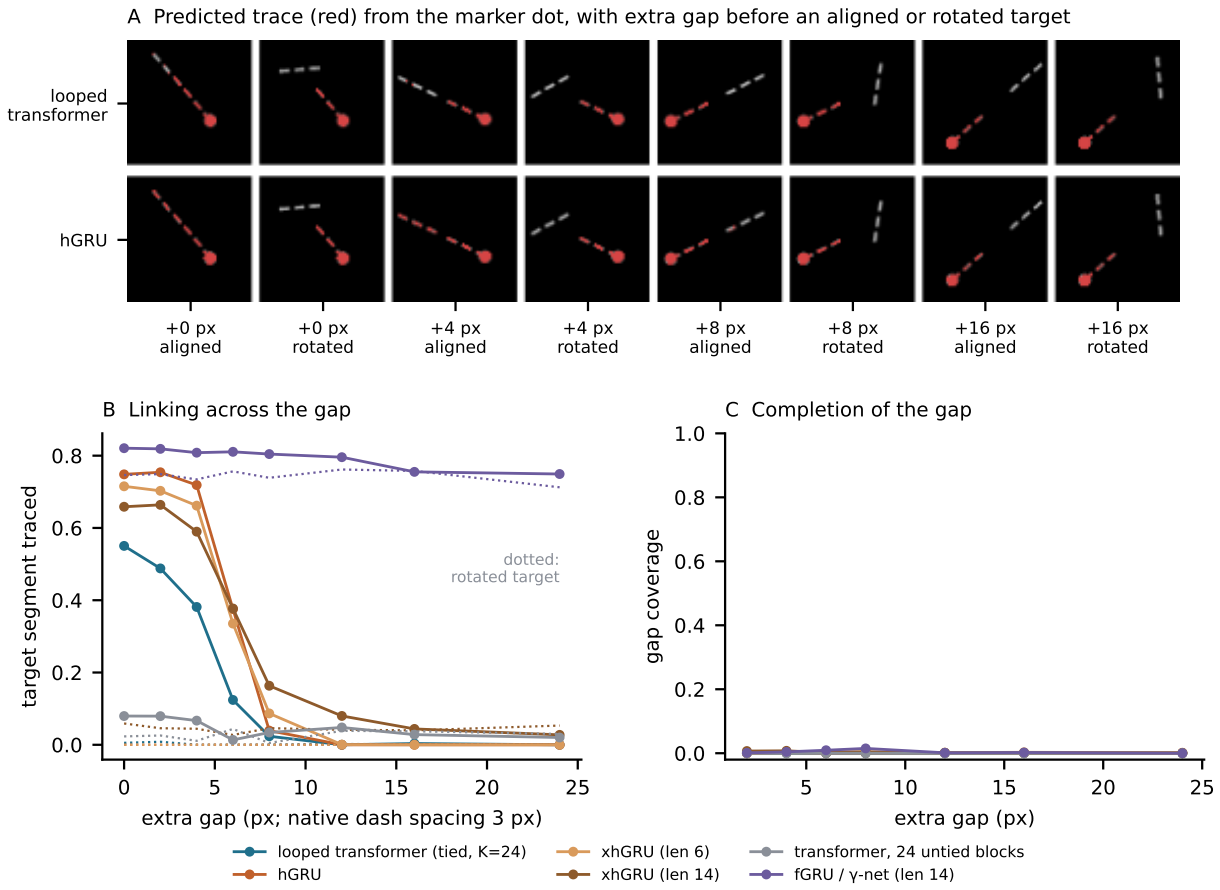



**Figure 10:** Contour-tracing networks (trained to segment the dashed contour a marker sits on) probed with a marked dashed segment, an extra gap, and a target segment that continues the path or is rotated in place. (A) Predicted traces. (B) Share of the target traced; solid, aligned; dotted, rotated. (C) Share of the missing contour in the gap predicted as contour.

iterations), fGRU/$\gamma$-net and weight-tied looped local-attention transformers trained to segment the dashed contour on which a marker sits, a Pathfinder-style curve-tracing task [20, 40] (validation IoU 0.92–0.996 on 128 px images; taken from a companion project without retraining; the training target is the full dashed contour, so a network that completes gaps has no reason to label them). Each saw a marker on a short dashed segment followed, along the same line or arc, by a second segment after an extra gap, or by the same segment rotated in place (segment-pair probes; Fig. 10). *Traced* is the share of the target segment's pixels labelled as part of the marked contour, and *gap coverage* the share of the missing contour labelled as contour (both with probability above 0.5). At the dash spacing of their training data, the tied transformer, the hGRU and the two xhGRUs extended the trace onto the aligned segment (mean 0.55–0.75 of its pixels) and almost never onto the rotated one (mean at most 0.06): they link by good continuation. For these four tracers, pooling aligned straight and curved segment pairs, linking halved after only 5–6 px of extra gap, averaged 0.02 at 12 px, and mean coverage of the missing contour stayed below 1% of gap pixels (at most 0.01; the fGRU reaches 1.5%). Running the looped transformer for twice its trained iterations reduced tracing of the aligned target at native spacing (0.55 to 0.41) and did not extend linking across larger gaps, an untied 24-block transformer of the same depth traced only 0.08 of the aligned target at

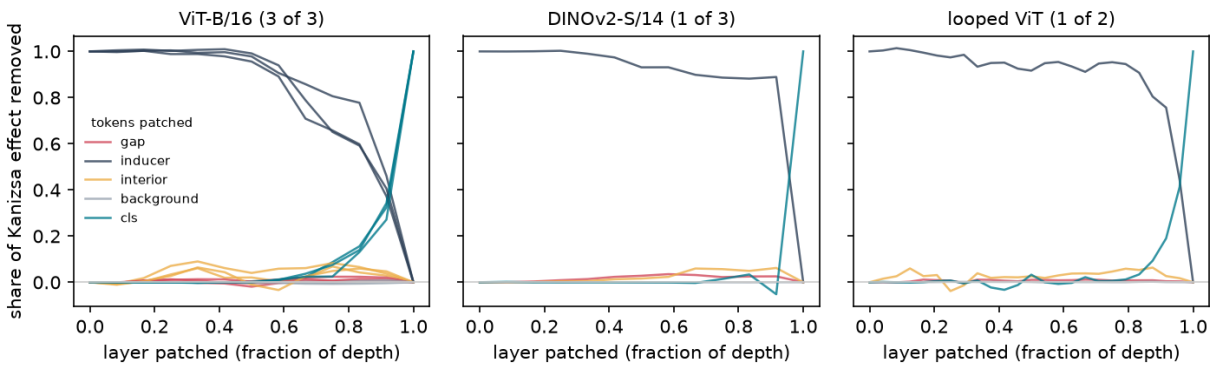



**Figure 11:** Activation patching in the transformers with a Kanizsa effect (log-odds of the true class at least 0.5 higher than with rotated inducers; inducer radius at least 1.5 patches). Share of that difference removed when one set of tokens at one block is replaced by the rotated-inducer display's (gap, inducer, interior and background tokens; class token), against the block's depth; one line per network.

native spacing, and the fGRU/$\gamma$-net labelled the rotated segment nearly as often as the aligned one (0.75 versus 0.82), so its spread is not contour-specific. In their segmentation output, networks that solve contour tracing therefore show the same profile as our classifiers: linking of present, aligned segments over short distances, and almost no labelling of absent contour; their training target did not require such labelling, so this does not rule out completion in their internal states.

### 4.8 The transformers' Kanizsa evidence does not pass through the gaps

The probes ask whether the empty cells represent an edge; activation patching [19, 32] asks whether the evidence the classifier uses passes through them. In a transformer every token reads every other at every block, so a Kanizsa figure could be recognised through the gap tokens even if no edge is represented there. For each Kanizsa display of the battery and its rotated-inducer control, which differ only in the inducer pixels, we ran the fine-tuned network on one display, replaced one set of tokens with the other display's at one block, or at that block and every later one, and measured the change in the log-odds of the true class. The sets are the gap tokens (cells crossed by the edge between neighbouring inducers that contain no inducer pixels in either display), the inducer tokens, the interior tokens, and as many background tokens outside the shape as there are gap tokens; we report the share of the Kanizsa-minus-rotated difference that the replacement removes.

A Kanizsa effect large enough to localise (at least 0.5 in log-odds) appears only with inducers of radius 1.5 patches or more, and there in 5 of the 12 transformers: the three ViT-B/16 seeds, one of three DINOv2-S seeds and the looped ViT-B/16 (accuracy 0.53–0.78 on these Kanizsa figures, 0.18–0.45 with rotated inducers); the others, all four MAE networks among them, reach at most 0.33. In these five networks, giving the gap tokens the rotated display's content at every block removes at most 0.05 of the effect, no more than the background tokens (at most 0.01), and the interior tokens remove 0.05–0.15; the inducer tokens carry all of it (1.00) until the last blocks, where it passes to the class token (Fig. 11). When the same gaps are closed by drawn lines (inducer radius up to 1 patch), the gap tokens carry 0.38–0.88 of the effect in all twelve networks, so the test finds evidence in the gaps where there is some. Where a Kanizsa effect exists the gaps are short, about six gap tokens per display against about eighty inducer tokens; within that limit, the transformers read their preference for aligned inducers from the inducers themselves, not from the space between them.

### 4.9 Circuits trained for incremental grouping label an illusory surface only across small gaps, and without needing recurrence

In incremental grouping, a label spreads from a cued point across the object it belongs to and stops at the object's boundary [39]. Mollard et al. [33] trained a recurrent $\gamma$-net [30] to segment the object at a cued location in natural images and found that the time its grouping signal takes to reach a point predicts human reaction times. No code or weights have been released, so we reimplemented and trained it from the paper's Methods: frozen Panoptic FPN ResNet-101 features [24], four recurrent levels of hGRU units with top-down connections, a $5 \times 5$ horizontal kernel at the finest level, 40 timesteps, and contractor recurrent back-propagation (C-RBP; 29) on COCO [27] (Appendix E). To find what the circuit's behaviour depends on, we also trained it with the reference code's penalty weight, with $3 \times 3$, $9 \times 9$ and $15 \times 15$ finest-level kernels, and with the fGRU cell of our other circuits in place of the hGRU (also on ViTDet ViT-B features, trained on COCO masks; 26); we replaced the recurrent hierarchy with one weight-tied transformer block looped for 40 steps, in which every position attends to every other at every step; and we trained single-step controls of the hGRU and of the looped transformer, which read out after one step. On cued COCO validation objects the 12 networks reach an IoU of 0.43–0.55 (Table 1; one $9 \times 9$ run, whose Lipschitz penalty rose late in training, 0.28), against 0.74 reported on the images used by Mollard et al. [33].

We placed the cue at the centre of each polygon of the battery ($n$ = 3–6, inducer radius 1–2 patches) and compared the region labelled as the cued object with the polygon. On the CNN backbone no network segments the Kanizsa figure (IoU at most 0.25), a thin real outline is segmented no better (at most 0.44), and most networks segment a mid-grey filled polygon (at least 0.48; Appendix E). At these proportions, then, the method cannot tell an illusory boundary from no boundary, because it does not register a drawn one either. We therefore used the large hexagon and varied the support ratio $s$, the share of each side bounded by contour pixels [45], from 0.50 to 0.95, which narrows the gap between neighbouring inducers from 36 to 4 px, and compared the Kanizsa display with rotated inducers, full discs at the same positions (which specify no polygon), thick corner segments, the same aligned inducers with each gap closed by a real solid or dashed line of the battery's outline width, and a blank page (Fig. 12). We report *fill*, the share of the hexagon's white pixels labelled as the cued object, beyond one patch from the cue, at the network's read-out step.

The looped transformer fills 0.85 of the hexagon with aligned inducers and 7 px gaps, as much as when the gaps are closed by a line (0.86), against 0.01 with rotated inducers, 0.00 with full discs and 0.02 with corner segments, and the label stays inside: just outside the gaps 0.08 is labelled, against 0.36 on a blank page, where the label spreads (0.63 of the hexagon's area; Figs. 12B, 13). This specificity holds only for small gaps. At 18 px a closing line is still filled (0.74) but the illusory gap is not (0.04), and at 29 px even the closing line is not (0.03). The circuits thus show three ranges (Fig. 12C): up to about 11 px, illusory gaps are filled like closed ones; from 14 to 22 px, closed gaps are filled and illusory ones are not; from 29 px, neither. All but two of the battery's twelve combinations of $n$ and inducer radius have gaps wider than 22 px.

Four controls locate what sets this limit. *Recurrence is not needed.* The single-step controls fill the Kanizsa hexagon as much as the looped transformer (looped transformer 0.79, hGRU 0.98 at 7 px) and stop at the same gaps (18 px: 0.22 and 0.29, against 0.56 and 0.94 with closing lines). What they lack is selectivity: they also label the hexagon between full discs (0.48 and 0.64), the hGRU between rotated inducers (0.31), and the looped transformer the whole blank page (1.00). Over 40 steps the looped transformer removes the fill between full discs (0.00) and reduces the spread on a blank page to 0.63 while keeping the aligned fill; in the fGRU trained with the weighted loss, by contrast, a label that fills the Kanizsa hexagon at step 40 (0.68; discs 0.19) floods the image by step 60 (Fig. 13). *The kernel's reach does not set the limit.* hGRUs with $3 \times 3$ and $15 \times 15$

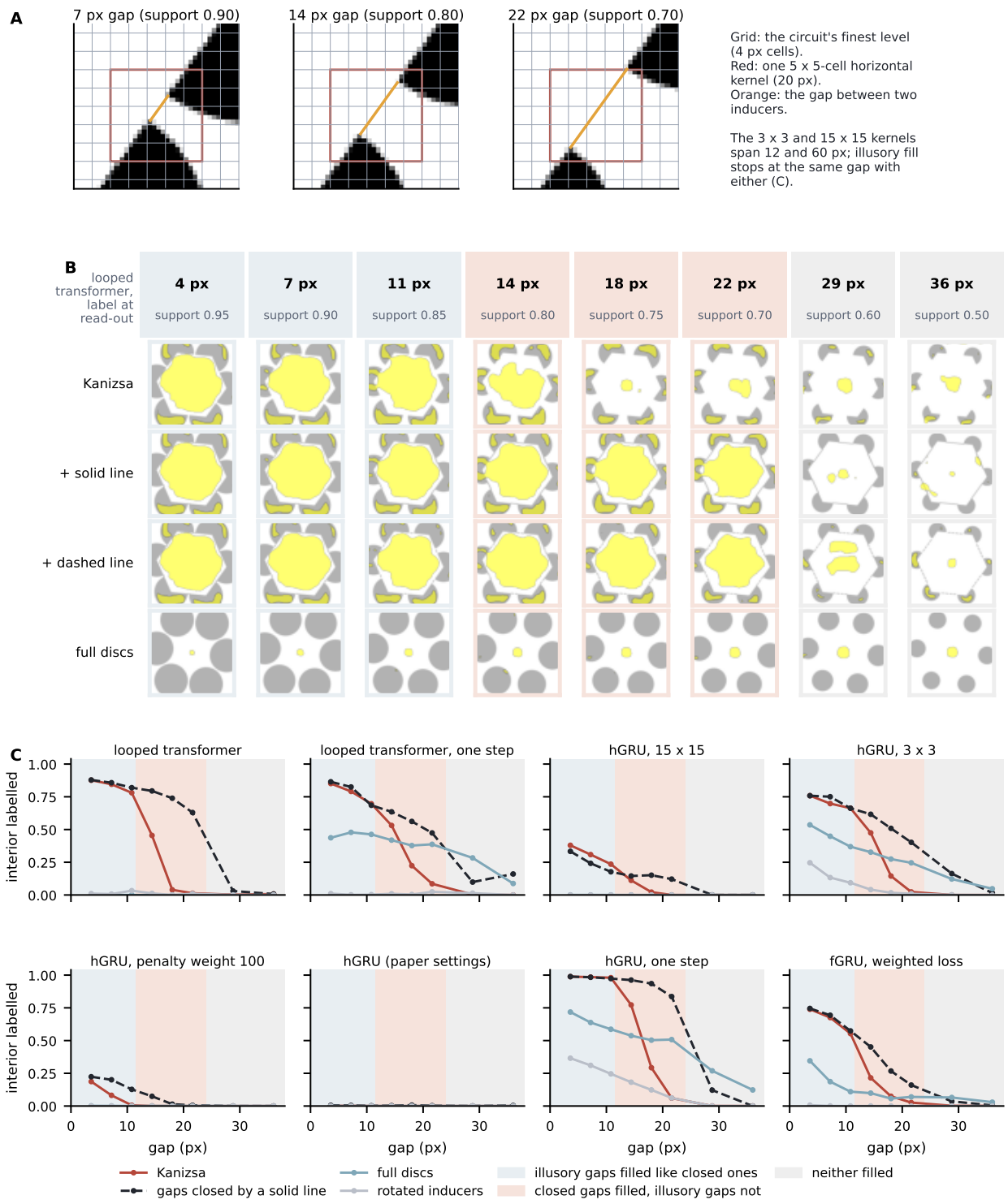



**Figure 12:** Cued segmentation circuits (after 33) on a large hexagon (circumradius 4.5 patches) at eight gap sizes, cue at the centre. (A) One gap enlarged on the 4 px grid of the circuits' finest recurrent level, with the 20 px reach of the paper's $5 \times 5$ horizontal kernel (red) and the gap (orange). (B) The looped transformer's label (yellow) at its read-out step for the Kanizsa display, the same inducers with the gaps closed by a solid or dashed line, and full discs at the same positions; header colours, the three gap ranges of (C). (C) Share of the hexagon's interior labelled against gap size (mean over four orientations) for eight circuits: Kanizsa, gaps closed by a solid line, full discs and rotated inducers. Shaded ranges follow the looped transformer: illusory gaps filled like closed ones; closed gaps filled and illusory gaps not; neither.

finest-level kernels (12 and 60 px) both stop between 14 and 18 px (18 px: 0.14 and 0.02, against 0.51 and 0.15 with closing lines), as the single-step controls do, consistent with the bridgeable gap's independence from the receptive field in the battery (Fig. 5). *The backbone does.* On ViTDet features the fGRU's fill does not depend on the gaps: at 29 px it labels 0.78 of the Kanizsa hexagon, where a closing line gets 0.51, and it fills between full discs (0.83 at 7 px), so it labels the region the inducers surround whether or not the gaps are closed. *Contrast polarity changes the outcome.* With white inducers on black, the looped transformer labels most of the hexagon whatever the inducers (Kanizsa 0.88, rotated 0.72, discs 0.59, blank 1.00); only containment remains: at 14 px, 0.09 of the strip just outside the sides is labelled with aligned inducers and 0.56 with rotated ones.

The circuit of Mollard et al. [33] varies most between runs. With the paper's settings it segments COCO objects (0.48) but labels nothing on any display, not even the filled hexagon (0.00), so it cannot be tested; with reversed contrast it labels the filled hexagon (0.65) but not the Kanizsa one (0.01). With the reference code's penalty weight it labels the filled hexagon (0.68) and the Kanizsa one only at the smallest gaps (0.08 at 7 px, closing line 0.20). The $3 \times 3$ and $15 \times 15$ runs, trained identically apart from the kernel, fill 0.70 and 0.31 at 7 px, the label growing over steps (gaps up to 11 px, $3 \times 3$: 0.15, 0.71 and 0.84 after 20, 40 and 60 steps); the $15 \times 15$ run leaves full discs and rotated inducers empty (0.00 and 0.00), the $3 \times 3$ run does not (discs 0.45). The $9 \times 9$ run labels nothing, also at iteration 12,000, before its penalty rose (filled hexagon 0.02). With one run per setting, these differences cannot be attributed to the settings. The looped transformer's selectivity is not reproducible either: a second run, identical apart from the random seed, labels the Kanizsa hexagon at 7 px gaps (0.96) but also the hexagon between rotated inducers (0.98) and full discs (0.98), and still 0.78 of the Kanizsa hexagon at 29 px; like the fGRU on ViTDet features, it labels the region the inducers surround (Table 1).

Trained on COCO alone, most of these circuits do not segment a thin drawn outline, so we also trained the looped transformer and the $15 \times 15$ hGRU, each with 40 steps and with one, on batches half of which were drawn polygons (filled, outlined, or outlined with inducers at the vertices, aligned or rotated; never a Kanizsa figure without its outline). We counted a polygon as seen when at least half of its interior and at most a tenth of a band just outside its sides are labelled, over $n = 3$–$8$ sides, four sizes, support ratios 0.50–0.90 and both polarities. These circuits see nearly every drawn outline (looped transformer 99%, hGRU 98%) and most Kanizsa figures (87% and 70%), but also many figures whose inducers are all rotated by 15–180° (27–57% and 9–37%; full discs 10% and 8%, blank page 0%); after one step they see rotated inducers more often (41–65% and 49–77%). Recurrence reduces this response to arrangements that specify no polygon without removing it.

Within the limits of single training runs of a reimplementation that differs from the original in how the cue enters, in its training images and in its accuracy, circuits trained to spread a cued label across objects label the surface of a Kanizsa figure, and in one of two runs not its controls, only where the gaps are a few pixels wide, under one patch of the battery; a single step does as much, any horizontal kernel from $3 \times 3$ to $15 \times 15$ stops at the same gap, and a ViT backbone fills regardless of the gaps. What bridges these gaps is the feedforward features; recurrence reduces, without removing, the label in regions that specify no shape. At the proportions of standard Kanizsa figures, these circuits treat the illusory surface as empty space.

### 4.10 Auditing published closure measures

We recreated each measure's stimuli from its published specification and computed it on trained networks and on null observers that cannot close contours: untrained networks (three seeds for the closure-measure audit, one for the extended configural-effect controls; the brightness audit uses no networks), a Gabor energy bank, raw pixels, and a *pixel-growth* observer that encodes only the




**Figure 13:** The cued label (yellow) step by step on the large hexagon with 7 px gaps (support ratio 0.90), cue (orange) at the centre. Left, Kanizsa display after $t$ steps; right, rotated inducers, full discs and the gaps closed by a solid line at the network's read-out step (40, or 1 for the single-step controls, which are trained and read out after one step).

**Table 1:** All trained cued segmentation circuits: IoU on cued COCO validation objects (all, and objects under 5% of the image), and the share of the large hexagon's interior labelled at the read-out step with 7 px gaps (filled hexagon, Kanizsa, rotated inducers, full discs, thick corner segments, gaps closed by a solid line, blank page) and with 18 px gaps (Kanizsa, solid line); mean over four orientations. Kernel sizes are those of the finest recurrent level. †Lipschitz penalty rose after iteration 12,000; two rollbacks. ‡Trained before the C-RBP corrections of Appendix E.

| | COCO IoU | | fill, 7 px gaps | | | | | | | fill, 18 px gaps | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| network | all | small | filled | Kanizsa | rotated | discs | corners | line | blank | Kanizsa | line |
| hGRU, $5 \times 5$ (paper settings) | 0.48 | 0.41 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 |
| hGRU, $5 \times 5$, penalty weight 100 | 0.50 | 0.42 | 0.68 | 0.08 | 0.00 | 0.00 | 0.00 | 0.20 | 0.00 | 0.00 | 0.01 |
| hGRU, $3 \times 3$ | 0.55 | 0.44 | 0.64 | 0.70 | 0.13 | 0.45 | 0.37 | 0.75 | 0.01 | 0.14 | 0.51 |
| hGRU, $9 \times 9$ † | 0.28 | 0.18 | 0.06 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 |
| hGRU, $15 \times 15$ | 0.54 | 0.44 | 0.54 | 0.31 | 0.00 | 0.00 | 0.19 | 0.24 | 0.00 | 0.02 | 0.15 |
| hGRU, $5 \times 5$, one step | 0.55 | 0.45 | 0.74 | 0.98 | 0.31 | 0.64 | 0.10 | 0.98 | 0.00 | 0.29 | 0.94 |
| fGRU (paper settings) ‡ | 0.43 | 0.35 | 0.28 | 0.28 | 0.01 | 0.10 | 0.00 | 0.34 | 0.00 | 0.01 | 0.27 |
| fGRU, weighted loss ‡ | 0.50 | 0.35 | 0.40 | 0.68 | 0.00 | 0.19 | 0.03 | 0.69 | 0.00 | 0.07 | 0.27 |
| fGRU on ViTDet ViT-B | 0.50 | 0.40 | 0.86 | 0.55 | 0.06 | 0.83 | 0.43 | 0.64 | 0.00 | 0.92 | 0.54 |
| looped transformer | 0.46 | 0.32 | 0.99 | 0.85 | 0.01 | 0.00 | 0.02 | 0.86 | 0.63 | 0.04 | 0.74 |
| looped transformer, second seed | 0.45 | 0.30 | 0.76 | 0.96 | 0.98 | 0.98 | 0.87 | 0.97 | 0.00 | 0.83 | 0.93 |
| looped transformer, one step | 0.51 | 0.38 | 0.99 | 0.79 | 0.00 | 0.48 | 0.66 | 0.83 | 1.00 | 0.22 | 0.56 |

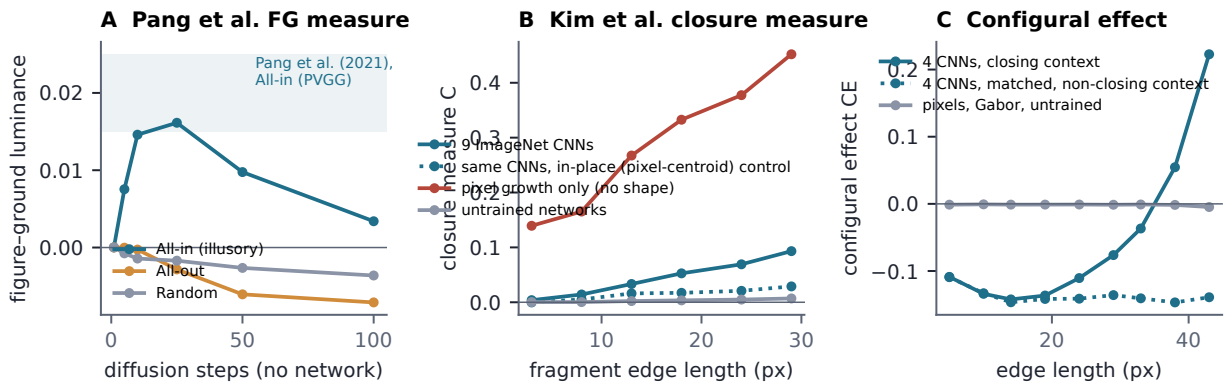



**Figure 14:** Published measures recomputed on null observers. (A) The figure–ground luminance difference of Pang et al. [35] computed on input images smoothed by diffusion; shaded band, magnitude reported for the All-in condition. (B) The closure measure $\bar{C}$ of Kim et al. [22] and Zhang et al. [54] for nine ImageNet CNNs, for the same CNNs with a disordered control that keeps contour pixels in place, for an observer that only encodes the radial distribution of contour pixels, and for untrained networks. (C) The configural effect (CE) of Zhang et al. [54] with a closing context and with the same components rotated so that no closed shape can form, against pixel, Gabor and untrained observers.

radial distribution of contour pixels about their centroid (our term; it differs from the pixel overlap that Kim et al. [22] control, which compares images pixel by pixel; Fig. 14).

**Illusory brightness [35].** The reported evidence is a figure-ground luminance difference (FG) of about 0.02 in reconstructions of Kanizsa squares whose inducers all face inward (All-in), which stays above zero after the first time step. Diffusing the input image produces an FG of +0.016 for All-in and -0.003 and -0.002 for the All-out and Random conditions after 25 steps, rising and then decaying as reported for their predictive-coding VGG (PVGG). A PCA reconstruction learned from natural images, with no feedback, gives +0.055 versus -0.032. The FG signature therefore does not require contour completion; it follows from smoothing inducers whose mass lies outside the illusory square.

**Closure measure [22, 54].** The closure measure is the cosine similarity of a network's representations of aligned fragments and a complete triangle minus that of disordered fragments (corners rotated about their vertices, as in 54) and the complete triangle:

$$\bar{C} = \mathbb{E}\big[\cos\big(f(x_{\text{aligned}}), f(x_{\text{complete}})\big) - \cos\big(f(x_{\text{disordered}}), f(x_{\text{complete}})\big)\big], \tag{1}$$

where $f$ is the representation at the layer read out, and the expectation runs over image triples in which the complete triangle differs from the fragments in orientation and position and shares their background [22]. Our replication reproduces the published pattern: trained CNNs score +0.019 to +0.087, rising with edge length, and untrained networks score +0.003. But the pixel-growth observer, which discards all angular information, scores +0.289 (+0.452 at the longest edges), several times any network. The disordered control rotates each corner about its vertex and moves the contour pixels outward; when we instead rotate each corner in place about the centroid of its own pixels, VGG16 falls from +0.029 to +0.005 and DenseNet-121 from +0.025 to +0.005, the two networks reported to show closure most consistently [54]. A residual of +0.013 to +0.028 remains in 7 of the 9 networks. The pairing rule also matters: Kim et al. [22] pair complete and fragment images that differ in global orientation and position, to ensure that there is no more pixel overlap between complete and aligned images than between complete and disordered images; when the paired images share orientation and position, raw pixels alone score +0.200 (versus +0.011 with

the published pairing), and the published version of Zhang et al. [54] does not state which pairing was used. Averaged over edge lengths, every trained network scores at most +0.014 on Kanizsa triangles, and ViT-B/16 scores +0.066 on line fragments.

**Configural effect [7, 54].** The network configural effect compares the distance between the representations of a base pair of images with that of a composite pair formed by adding the same components, the context, to both: $\mathrm{CE} = (D_{\mathrm{composite}} - D_{\mathrm{base}})/(D_{\mathrm{composite}} + D_{\mathrm{base}})$ with $D$ the Euclidean distance between representations, positive for a configural superiority effect and negative for a configural inferiority effect. In the line-segment condition of Zhang et al. [54] the context closes a square; we call it the closing context. This measure survives the null observers. Pixels, Gabor filters and untrained networks score near zero, and with components long enough that the gaps are under half a patch, ResNet50 and DenseNet-121 reach +0.30 and +0.29. Replacing the context with the same components rotated outward, so that no closed shape can form, reverses the effect (-0.17 and -0.18). Three further controls show what it measures (stimuli in Fig. 15; results in Fig. 16A–C). *Arrangement.* A context that keeps every component's position and reduces the radial differences between arrangements (components rotated in place about the centroids of their own pixels) is exceeded by the closing context by only +0.01 at an edge length of 5 px, and by +0.24 at 43 px; a fully closed square gives +0.23, on average no more than a nearly closed one (the closing-context CE itself still rises from 43 to 47.5 px in eight of the ten networks, on average from 0.12 to 0.13; it is the closing-minus-in-place contrast that stops growing), so CE grows as the outline approaches completion and the closing contrast gains nothing further, on average, from closing it. The pixel-growth observer shows no such advantage of the closing over the in-place context at 43 px (+0.01; +0.01 at 5 px, and negative at intermediate lengths), so this observer does not account for the large contrast at 43 px, although centroid matching does not equalize the radial histograms exactly. The effect thus tracks how much of the outline is physically present, not gaps to be bridged. Averaged over all conditions, the ten trained networks (nine CNNs and ViT-B/16) score -0.16, mostly a configural inferiority effect, and CE is positive by a one-sample $t$-test across stimulus sets (nominal $p < 0.05$, uncorrected) in only 22% of network–edge cells with a closing context (0% in place). The pixel-growth observer scores -0.36 on CE itself, so radial pixel statistics alone can produce a configural inferiority effect of this size; the arrangement contrast above is what separates the networks from it. *Inducer shape.* Five notched inducers (notched disc, notched square, hexagon, cross and ring) have identical notch edges and hence the same ratio of luminance-specified to total edge length, which predicts the clarity of human illusory boundaries [45], but a network's CE varies across them by a median of 0.16 and changes sign in 5 of 10 networks (ResNet50: +0.03 with notched discs, +0.31 with notched squares). *Inducer size.* Enlarging the inducer disc while keeping the notch edges fixed, which leaves that ratio unchanged, moves CE from -0.18 to -0.11, while the closing context's advantage over the in-place control stays positive (at least +0.02). CE is therefore specific to closing arrangements of nearly complete outlines, but it is not invariant to manipulations that leave that ratio unchanged.

**Accuracy under edge removal [53].** This protocol already concludes that CNNs lack closure. Because corners are kept, it can be passed by reading corner angles (see above); because gaps scale with edge length, gap size in patches is confounded with the number of sides; and because, in their released stimulus code, test polygons are smaller than training polygons (radius 60 versus 67 px), the reported drop at 10% removal mixes size change with removal.

### 4.11 Saliency overlays are often not faithful

Overlays are a common argument that a network "sees" a triangle. We scored each map with a deletion test that removes contour pixels in map order or in random order [36], and call the resulting

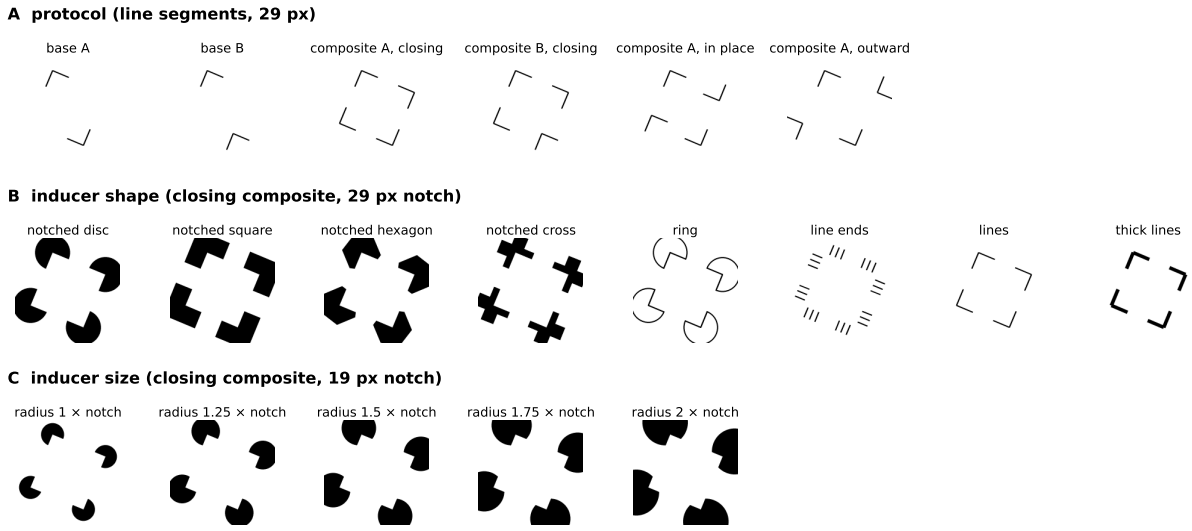



**Figure 15:** The configural-effect stimuli. (A) The protocol with line segments (edge length 29 px): the base pair differs in the orientation of one corner (flipped about the centroid of its own pixels); the composite pair adds the same two corners, the context, to both. The closing context completes the square; the in-place context rotates each added corner by 90° about the centroid of its own pixels; the outward context rotates it by 180° about its vertex. (B) The eight inducers of the shape audit, each specifying the same corner of the same square with the closing context (29 px notch); the first five have identical notch edges. (C) The size audit: notched discs enlarged at a fixed 19 px notch, so the notch edges and the implied square are unchanged.

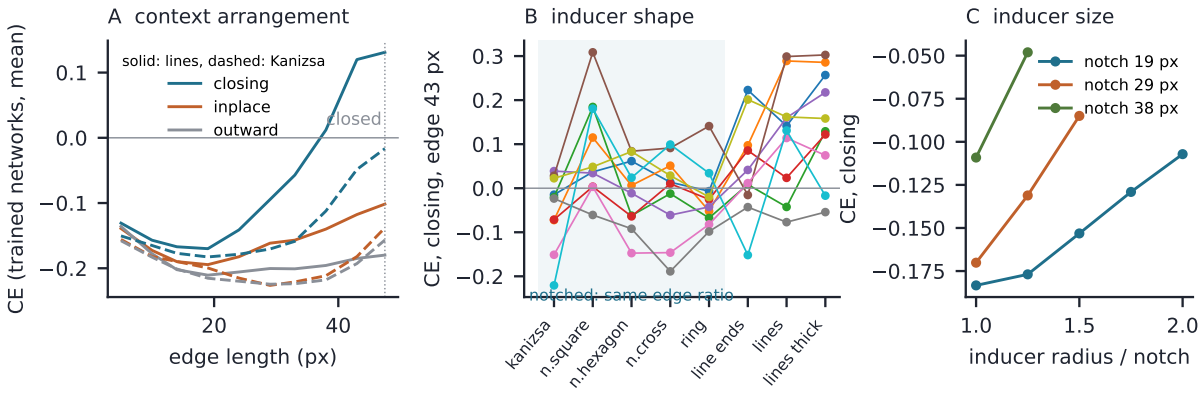



**Figure 16:** (A) Configural effect for the ten trained networks (nine CNNs and ViT-B/16) with closing, in-place (component-centroid-matched, non-closing) and outward contexts against edge length; 47.5 px is a closed square. (B) CE per network across inducer shapes; shaded, notched inducers with identical notch edges (same specified-to-total edge ratio). (C) CE against inducer radius at a fixed notch.

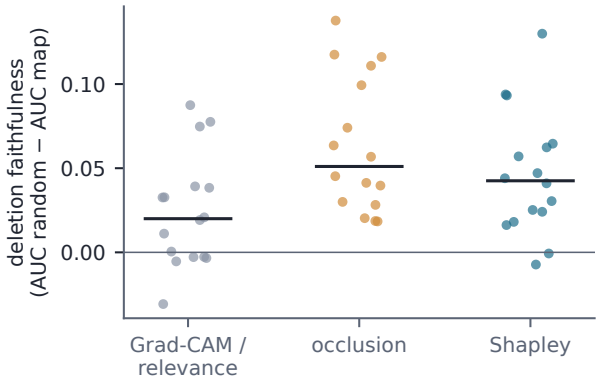



**Figure 17:** Deletion faithfulness of three attribution methods across networks (one point per network; bar, median).

score faithfulness, in the sense in which Selvaraju et al. [43] use the word (Appendix B). Grad-CAM [43] and class-specific relevance maps [9] have a median faithfulness of +0.020 and are near zero ($|\cdot| < 0.02$) for 7 of 16 networks; occlusion [52] and Shapley values computed over the stimulus's own pieces score +0.051 and +0.043 (Fig. 17). This is separate evidence from the sanity checks of Adebayo et al. [2], which Grad-CAM passed and which test a different property.

## 5 Discussion

Across 24 networks and four published measures, we find at most a weak illusory-contour signal, consistent across seeds only in ViT-B/16 (Kanizsa accuracy at most 0.39 against 1.00 on complete outlines), and the published illusory-brightness and representational-similarity signatures of closure are reproduced by observers that cannot close contours. What networks do show is integration of nearby, consistently oriented contour pieces: fine-tuned transformers use both the positions and the orientations of fragments, and ImageNet CNNs show a configural effect whose advantage for closing arrangements grows as the outline approaches completion. In CNN and recurrent networks the gaps that halve closure are, with one exception (two counting single-iteration controls), smaller than the measured effective receptive field, so the measured receptive field does not appear to be what limits closure. This places current networks closer to the local association-field account of contour integration [13] than to Gestalt closure: they link pieces that are near and aligned, and do not complete what is absent. In the terms of Roelfsema [39], the networks perform base grouping, by units tuned to conjunctions of nearby aligned elements, but show no sign of incremental grouping, in which activity spreads along an object through horizontal and feedback connections. Nor do they show the growing reach that incremental grouping predicts [37]: the gaps they bridge stay within the receptive field, and neither iterations nor depth extend them. A related dissociation has been reported for vision-language models, which give the expected answer to classic illusions but keep it when the inducers are reversed [46]; recognizing an illusory configuration and being susceptible to it are different properties, and our measures target the second. Networks trained to trace contours, the task recurrence is thought to be needed for, share this profile: they link aligned pieces a few pixels beyond their training statistics and label almost none of the missing contour as contour. Recurrence does not change this picture. In some fGRU circuits the gap similarity rises over iterations, but classifiers trained on complete outlines have little reason to use it, and single-iteration controls perform as well in all but one case.

**Seeing the inducers is not enough.** A network's feature extractors respond where there are features; between the inducers there are none, and nothing we measured puts a response there. Wherever the read-outs work, a contour probe and the most contour-selective units read a real line at 6% contrast almost fully, yet value the illusory edge at no more than 3.1% of a real line's contrast (Section 4.5). This holds at every layer, including late transformer blocks whose tokens attend to every inducer and convolutional and recurrent networks whose receptive fields span the gap, and it holds when the inducers are familiar from training (Appendix C); missing sides stay empty however many others are drawn. A receptive field large enough to contain all the inducers gives a unit access to them, but nothing in training for recognition makes it respond where the image is empty. This is not a necessary property of visual systems: neurons in V1 respond more inside a figure than outside it when the local image is identical [25], a modulation carried by recurrent and feedback processing [23, 42]. In incremental grouping, however, the spreading label enhances responses the image already drives [8, 39], so even a network that implements it would have nothing to label between the inducers. Circuits trained for such spreading confirm this: reimplementations of the GammaNet of Mollard et al. [33] label the interior of a Kanizsa figure only across gaps of a few pixels, as much after one step as after 40, and recurrence only reduces, without removing, the label in regions that specify no shape, such as those between full discs or rotated inducers (Section 4.9).

**Recommendations.** Closure tests should (i) specify gaps in patch or receptive-field units; (ii) remove corners or otherwise prevent corner reading; (iii) include controls that keep the contour pixels, their pixel growth and their orientations while breaking the arrangement; and (iv) report null observers (untrained networks, a filter bank, pixels and a pixel-growth observer) alongside trained ones.

**Limitations.** ViT-B/16 and DINOv2-S were trained with three seeds and ResNet50 with five, whose fragment and arrangement effects vary by up to 0.03 and 0.05 (s.d.); the others were trained once, so differences of that size between them remain uncertain without repeated training. Published stimuli were recreated from their papers' descriptions rather than original code. We did not test the implicit-prior account of Toosi and Miller [49], or abutting-grating illusions, and we did not collect human data on our stimuli. Pathfinder pretraining [28] of our recurrent circuits, and of a ViT-B/16, remained at chance after 1–2 h of training per network; networks from a companion project that were trained to trace contours stand in for it (Fig. 10).

## Use of AI tools

Code for stimulus generation, training, evaluation and figures, and drafts of this manuscript, were produced with the assistance of a large language model (Claude, Anthropic). All experimental design decisions, results and text were reviewed by the authors.

## Code and data

Stimulus generator, training and evaluation code, and per-image results will be released with the final version.

## References

[1] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4190–4197, 2020. doi: 10.18653/v1/2020.acl-main.385.

[2] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[3] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations (ICLR), Workshop Track*, 2017. URL https://openreview.net/forum?id=HJ4-rAVtl.

[4] Nicholas Baker, Gennady Erlikhman, Philip J. Kellman, and Hongjing Lu. Deep convolutional networks do not perceive illusory contours. In *Proceedings of the 40th Annual Conference of the Cognitive Science Society*, 2018.

[5] Johannes Ballé, Valero Laparra, and Eero P. Simoncelli. Density modeling of images using a generalized normalization transformation. In *International Conference on Learning Representations*, 2016.

[6] Irving Biederman. Recognition-by-components: A theory of human image understanding. *Psychological Review*, 94(2):115–147, 1987. doi: 10.1037/0033-295X.94.2.115.

[7] Valerio Biscione and Jeffrey S. Bowers. Mixed evidence for Gestalt grouping in deep neural networks. *Computational Brain & Behavior*, 6(3):438–456, 2023.

[8] Tobias Brosch, Heiko Neumann, and Pieter R. Roelfsema. Reinforcement learning of linking and tracing contours in recurrent neural networks. *PLOS Computational Biology*, 11(10):e1004489, 2015. doi: 10.1371/journal.pcbi.1004489.

[9] Hila Chefer, Shir Gur, and Lior Wolf. Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 387–396. IEEE, 2021. doi: 10.1109/ICCV48922.2021.00045.

[10] Hila Chefer, Shir Gur, and Lior Wolf. Transformer interpretability beyond attention visualization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 782–791, 2021. doi: 10.1109/CVPR46437.2021.00084.

[11] Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. Universal transformers. In *International Conference on Learning Representations*, 2019.

[12] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

[13] David J. Field, Anthony Hayes, and Robert F. Hess. Contour integration by the human visual system: Evidence for a local "association field". *Vision Research*, 33(2):173–193, 1993. doi: 10.1016/0042-6989(93)90156-Q.

[14] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In *Advances in Neural Information Processing Systems*, volume 38, pages 46242–46293, 2025. doi: 10.52202/085713-1380. arXiv:2502.05171.

[15] Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://openreview.net/forum?id=Bygh9j09KX.

[16] Angeliki Giannou, Shashank Rajput, Jy-Yong Sohn, Kangwook Lee, Jason D. Lee, and Dimitris Papailiopoulos. Looped transformers as programmable computers. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 11398–11442. PMLR, 2023.

[17] Michel Grabisch and Marc Roubens. An axiomatic approach to the concept of interaction among players in cooperative games. *International Journal of Game Theory*, 28(4):547–565, 1999. doi: 10.1007/s001820050125.

[18] David J. Heeger. Normalization of cell responses in cat striate cortex. *Visual Neuroscience*, 9 (2):181–197, 1992. doi: 10.1017/S0952523800009640.

[19] Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.

[20] Pierre Jolicoeur, Shimon Ullman, and Marilynn Mackay. Curve tracing: A possible basic operation in the perception of spatial relations. *Memory & Cognition*, 14(2):129–140, 1986. doi: 10.3758/BF03198373.

[21] Gaetano Kanizsa. *Organization in Vision: Essays on Gestalt Perception*. Praeger, 1979.

[22] Been Kim, Emily Reif, Martin Wattenberg, Samy Bengio, and Michael C. Mozer. Neural networks trained on natural scenes exhibit gestalt closure. *Computational Brain & Behavior*, 4 (3):251–263, 2021. arXiv:1903.01069.

[23] Lisa Kirchberger, Sreedeep Mukherjee, Ulf H. Schnabel, Enny H. van Beest, Areg Barsegyan, Christiaan N. Levelt, J. Alexander Heimel, Jeannette A. M. Lorteije, Chris van der Togt, Matthew W. Self, and Pieter R. Roelfsema. The essential role of recurrent processing for figure-ground perception in mice. *Science Advances*, 7(27):eabe1833, 2021. doi: 10.1126/sciadv.abe1833.

[24] Alexander Kirillov, Ross Girshick, Kaiming He, and Piotr Dollár. Panoptic feature pyramid networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6399–6408, 2019.

[25] Victor A. F. Lamme. The neurophysiology of figure-ground segregation in primary visual cortex. *The Journal of Neuroscience*, 15(2):1605–1615, 1995. doi: 10.1523/JNEUROSCI.15-02-01605.1995.

[26] Yanghao Li, Hanzi Mao, Ross Girshick, and Kaiming He. Exploring plain vision transformer backbones for object detection. In *Proceedings of the European Conference on Computer Vision*, 2022. doi: 10.1007/978-3-031-20077-9_17.

[27] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision*, pages 740–755, 2014.

[28] Drew Linsley, Junkyung Kim, Vijay Veerabadran, Charles Windolf, and Thomas Serre. Learning long-range spatial dependencies with horizontal gated recurrent units. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[29] Drew Linsley, Alekh Karkada Ashok, Lakshmi Narasimhan Govindarajan, Rex Liu, and Thomas Serre. Stable and expressive recurrent vision models. In *Advances in Neural Information Processing Systems*, volume 33, pages 10456–10467, 2020.

[30] Drew Linsley, Junkyung Kim, Alekh Ashok, and Thomas Serre. Recurrent neural circuits for contour detection. In *International Conference on Learning Representations*, 2020.

[31] Wenjie Luo, Yujia Li, Raquel Urtasun, and Richard Zemel. Understanding the effective receptive field in deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

[32] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[33] Sami Mollard, Alekh K. Ashok, Lore Goetschalckx, Drew Linsley, Thomas Serre, Sander M. Bohte, and Pieter R. Roelfsema. Recurrent network dynamics explain the time course of perceptual grouping in natural scenes. *arXiv preprint arXiv:2610.05419*, 2026.

[34] Michael C. Mozer, Shoaib Ahmed Siddiqui, Danny Sawyer, Sunny Sanyal, and Rosanne Liu. Recirculation. *arXiv preprint arXiv:2608.17981*, 2026.

[35] Zhaoyang Pang, Callum Biggs O'May, Bhavin Choksi, and Rufin VanRullen. Predictive coding feedback results in perceived illusory contours in a recurrent neural network. *Neural Networks*, 144:164–175, 2021. doi: 10.1016/j.neunet.2021.08.024.

[36] Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. In *British Machine Vision Conference*, 2018.

[37] Arezoo Pooresmaeili and Pieter R. Roelfsema. A growth-cone model for the spread of object-based attention during contour grouping. *Current Biology*, 24(24):2869–2877, 2014. doi: 10.1016/j.cub.2014.10.007.

[38] Dario L. Ringach and Robert Shapley. Spatial and temporal properties of illusory contours and amodal boundary completion. *Vision Research*, 36(19):3037–3050, 1996. doi: 10.1016/0042-6989(96)00062-4.

[39] Pieter R. Roelfsema. Cortical algorithms for perceptual grouping. *Annual Review of Neuroscience*, 29(1):203–227, 2006. doi: 10.1146/annurev.neuro.29.051605.112939.

[40] Pieter R. Roelfsema, Victor A. F. Lamme, and Henk Spekreijse. Object-based attention in the primary visual cortex of the macaque monkey. *Nature*, 395(6700):376–381, 1998. doi: 10.1038/26475.

[41] Martin Schrimpf, Jonas Kubilius, Michael J. Lee, N. Apurva Ratan Murty, Robert Ajemian, and James J. DiCarlo. Integrative benchmarking to advance neurally mechanistic models of human intelligence. *Neuron*, 108(3):413–423, 2020. doi: 10.1016/j.neuron.2020.07.040.

[42] Matthew W. Self, Roxana N. Kooijmans, Hans Supèr, Victor A. F. Lamme, and Pieter R. Roelfsema. Different glutamate receptors convey feedforward and recurrent processing in macaque V1. *Proceedings of the National Academy of Sciences*, 109(27):11031–11036, 2012. doi: 10.1073/pnas.1119527109.

[43] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision*, pages 618–626, 2017. doi: 10.1109/ICCV.2017.74.

[44] Lloyd S. Shapley. A value for $n$-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games, Volume II*, number 28 in Annals of Mathematics Studies, pages 307–318. Princeton University Press, Princeton, NJ, 1953. doi: 10.1515/9781400881970-018.

[45] Thomas F. Shipley and Philip J. Kellman. Strength of visual interpolation depends on the ratio of physically specified to total edge length. *Perception & Psychophysics*, 52(1):97–106, 1992. doi: 10.3758/BF03206762.

[46] Xiaoxiao Sun, Mingyang Li, Kun Yuan, Min Woo Sun, Mark Endo, Shengguang Wu, Changlin Li, Yuhui Zhang, Zeyu Wang, and Serena Yeung-Levy. Do VLMs perceive or recall? Probing visual perception vs. memory with classic visual illusions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25861–25870, 2026.

[47] Corentin Tallec and Yann Ollivier. Can recurrent neural networks warp time? In *International Conference on Learning Representations (ICLR)*, 2018. URL https://openreview.net/forum?id=SJcKhk-Ab.

[48] Leonard Tang and Dan Ley. Degraded polygons raise fundamental questions of neural network perception. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 9695–9706. Curran Associates, Inc., 2023. doi: 10.52202/075280-0425. Datasets and Benchmarks Track.

[49] Tahereh Toosi and Kenneth D. Miller. Emergence of illusory contours in robust deep neural networks by accumulation of implicit priors. *Journal of Vision*, 24(10):639, 2024.

[50] Johan Wagemans, James H. Elder, Michael Kubovy, Stephen E. Palmer, Mary A. Peterson, Manish Singh, and Rüdiger von der Heydt. A century of Gestalt psychology in visual perception: I. perceptual grouping and figure–ground organization. *Psychological Bulletin*, 138(6):1172–1217, 2012. doi: 10.1037/a0029333.

[51] Yuxin Wu and Kaiming He. Group normalization. In Vittorio Ferrari, Martial Hebert, Cristian Sminchisescu, and Yair Weiss, editors, *Computer Vision – ECCV 2018*, volume 11217 of *Lecture Notes in Computer Science*, pages 3–19, Cham, 2018. Springer International Publishing. doi: 10.1007/978-3-030-01261-8_1.

[52] Matthew D. Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In David Fleet, Tomas Pajdla, Bernt Schiele, and Tinne Tuytelaars, editors, *Computer Vision – ECCV 2014*, volume 8689 of *Lecture Notes in Computer Science*, pages 818–833, Cham, 2014. Springer International Publishing. doi: 10.1007/978-3-319-10590-1_53.

[53] Yuyan Zhang, Derya Soydaner, Fatemeh Behrad, Lisa Koßmann, and Johan Wagemans. Investigating the Gestalt principle of closure in deep convolutional neural networks. In *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*, pages 679–684, 2024. doi: 10.14428/esann/2024.es2024-111. arXiv:2411.00627.

[54] Yuyan Zhang, Derya Soydaner, Lisa Koßmann, Fatemeh Behrad, and Johan Wagemans. Finding closure: A closer look at the Gestalt law of closure in convolutional neural networks. *Computational Brain & Behavior*, 9(1):104–116, 2025. doi: 10.1007/s42113-025-00251-x. Preprint arXiv:2408.12460 (2024).

# A Results for every network

| Network | complete | frag. | CI frag. | scrambled | conflict: pos. | conflict: ori. | Kanizsa | CI Kan. | Brain-Score* |
|---|---|---|---|---|---|---|---|---|---|
| VGG16 (frozen) | 0.99 | 0.25 | +0.14 | 0.10 | 0.10 | 0.00 | 0.14 | +0.02 | 0.37 |
| ResNet50 (frozen) | 0.97 | 0.41 | +0.27 | 0.12 | 0.15 | 0.04 | 0.20 | -0.00 | 0.38 |
| ViT-B/16 (frozen) | 1.00 | 0.62 | +0.43 | 0.26 | 0.12 | 0.16 | 0.29 | +0.10 | – |
| DINOv2-S (frozen) | 0.97 | 0.63 | +0.51 | 0.20 | 0.02 | 0.08 | 0.19 | +0.00 | 0.35 |
| ResNet50 | 1.00 | 0.66 | +0.53 | 0.14 | 0.27 | 0.15 | 0.18 | +0.04 | 0.34 |
| ViT-B/16 | 1.00 | 0.82 | +0.60 | 0.27 | 0.05 | 0.18 | 0.29 | +0.08 | – |
| DINOv2-S | 1.00 | 0.94 | +0.79 | 0.30 | 0.03 | 0.33 | 0.25 | +0.10 | 0.34 |
| Looped ViT-B/16 | 1.00 | 0.81 | +0.60 | 0.39 | 0.04 | 0.42 | 0.39 | +0.09 | – |
| Looped DINOv2-S | 1.00 | 0.97 | +0.80 | 0.36 | 0.09 | 0.30 | 0.20 | +0.03 | 0.31 |
| fGRU-H | 1.00 | 0.32 | +0.13 | 0.12 | 0.14 | 0.12 | 0.15 | -0.00 | – |
| fGRU-H (T=1) | 1.00 | 0.30 | +0.18 | 0.18 | 0.09 | 0.11 | 0.14 | +0.00 | – |
| fGRU-H, 1x1 readout | 1.00 | 0.78 | +0.63 | 0.68 | 0.05 | 0.70 | 0.07 | -0.00 | – |
| $\gamma$-net | 1.00 | 0.58 | +0.36 | 0.41 | 0.08 | 0.52 | 0.04 | +0.00 | – |
| $\gamma$-net (T=1) | 1.00 | 0.68 | +0.50 | 0.56 | 0.09 | 0.53 | 0.12 | +0.04 | – |
| deep fGRU-H D8 | 1.00 | 0.54 | +0.36 | 0.11 | 0.11 | 0.00 | 0.16 | +0.06 | – |
| deep fGRU-H D8 (T=1) | 1.00 | 0.51 | +0.36 | 0.10 | 0.10 | 0.00 | 0.11 | +0.03 | – |
| deep fGRU D4 | 1.00 | 0.55 | +0.42 | 0.10 | 0.16 | 0.00 | 0.17 | +0.06 | – |
| deep fGRU D4 (T=1) | 1.00 | 0.51 | +0.34 | 0.15 | 0.10 | 0.03 | 0.18 | +0.03 | – |
| DN fGRU stack L4 | 1.00 | 0.51 | +0.29 | 0.24 | 0.15 | 0.12 | 0.13 | -0.01 | – |
| DN fGRU stack L4 (T=1) | 1.00 | 0.49 | +0.30 | 0.24 | 0.01 | 0.16 | 0.09 | +0.01 | – |
| DN fGRU stack L8 | 1.00 | 0.41 | +0.26 | 0.29 | 0.12 | 0.10 | 0.10 | -0.00 | – |
| DN fGRU stack L8 (T=1) | 1.00 | 0.61 | +0.40 | 0.40 | 0.01 | 0.21 | 0.11 | -0.00 | – |
| VGG16 (Zhang data)† | 0.29 | 0.19 | +0.06 | 0.12 | 0.11 | 0.01 | 0.17 | +0.00 | – |
| ResNet50 (Zhang data)† | 0.39 | 0.28 | +0.15 | 0.13 | 0.13 | 0.03 | 0.10 | +0.00 | – |

**Table 2:** All networks. complete: accuracy on complete outlines; frag. and CI: accuracy and closure index for corner-removed fragments of the main battery; scrambled and cue conflict: the orientation-control battery (a different set of fragments, so scrambled cannot be subtracted from frag.); Kanizsa: accuracy and closure index; Brain-Score*: mean of public-split V1, V2, V4, IT and behavioural benchmarks run locally [41] (not the leaderboard score; computed for a subset of networks, "–" otherwise). †Trained on the stimuli of Zhang et al. [53] (validation accuracy 0.88 and 0.85 on those stimuli); they transfer poorly to our battery.

# B Implementation details

Training used AdamW with cosine decay and 5% warm-up, label smoothing 0.1 and bf16 autocast (full precision for the DN stacks, whose eight-layer stack falls from 1.00 to 0.44 on complete outlines when evaluated in bf16); backbone learning rates were $10^{-5}$ (transformers) or $10^{-4}$ (CNNs and recurrent front ends), and new modules used $10^{-3}$ ($10^{-4}$ for the looped-transformer injection). fGRU cells used $9 \times 9$ horizontal kernels at stride 4 and kernels of 7, 5 and 3 at deeper stages; gate biases followed the chrono initialization [47] and horizontal kernels were orthogonally initialized

| Network | family | $\mathrm{ERF}_{50}$ or $P$ (px) | $g^*$ (px) | $g^*/(\mathrm{ERF}_{50}$ or $P)$ |
|---|---|---|---|---|
| VGG16 (frozen) | CNN | 38 | 6 | 0.16 |
| ResNet50 | CNN | 60 | 20 | 0.34 |
| ResNet50 (frozen) | CNN | 66 | 14 | 0.22 |
| deep fGRU-H D8 (T=1) | $T$=1 control | 16 | 9 | 0.58 |
| fGRU-H (T=1) | $T$=1 control | 19 | 15 | 0.78 |
| $\gamma$-net (T=1) | $T$=1 control | 35 | 39 | 1.09 |
| DN fGRU L4 (T=1) | $T$=1 control | 43 | 11 | 0.25 |
| DN fGRU L8 (T=1) | $T$=1 control | 60 | 51 | 0.84 |
| deep fGRU D4 (T=1) | $T$=1 control | 65 | 38 | 0.59 |
| Looped ViT-B/16 | ViT (patch) | 16 | 54 | 3.39 |
| ViT-B/16 (frozen) | ViT (patch) | 16 | 55 | 3.45 |
| DINOv2-S | ViT (patch) | 14 | $\geq$56 | $\geq$4.00 |
| ViT-B/16 | ViT (patch) | 16 | $\geq$64 | $\geq$4.00 |
| DINOv2-S (frozen) | ViT (patch) | 14 | 51 | 3.64 |
| Looped DINOv2-S | ViT (patch) | 14 | $\geq$56 | $\geq$4.00 |
| deep fGRU-H D8 | recurrent | 16 | 10 | 0.62 |
| fGRU-H, $1 \times 1$ readout | recurrent | 18 | 50 | 2.82 |
| fGRU-H | recurrent | 18 | 15 | 0.82 |
| $\gamma$-net | recurrent | 59 | 29 | 0.49 |
| deep fGRU D4 | recurrent | 70 | 27 | 0.39 |
| DN fGRU L4 | recurrent + DN | 58 | 17 | 0.30 |
| DN fGRU L8 | recurrent + DN | 74 | 44 | 0.60 |

**Table 3:** Critical gap $g^*$ against $\mathrm{ERF}_{50}$ (patch size $P$ for transformers) for every network, including the recurrent variants and single-iteration controls not drawn in Fig. 5.

| Network | seeds | CI corner | CI mid | arrangement | CI Kanizsa | Kanizsa range |
|---|---|---|---|---|---|---|
| ViT-B/16 | 3 | +0.62 ± 0.03 | +0.42 ± 0.04 | +0.54 ± 0.05 | +0.11 ± 0.03 | +0.08 to +0.14 |
| DINOv2-S | 3 | +0.78 ± 0.01 | +0.43 ± 0.01 | +0.64 ± 0.02 | +0.09 ± 0.07 | +0.01 to +0.15 |
| ResNet50 | 5 | +0.52 ± 0.03 | +0.23 ± 0.06 | +0.51 ± 0.03 | +0.02 ± 0.05 | -0.05 to +0.07 |

**Table 4:** Repeated training seeds: mean ± s.d. of the closure indices and the arrangement effect.

and then made symmetric between channels; $\alpha$ is kept positive by a softplus, and a top-down fGRU adds its output to the lower layer. Relevance maps followed Chefer et al. [9]: for each block, $\bar{A} = \mathbb{E}_h[(\nabla A \odot A)^+]$ with $\nabla A$ the gradient of the class logit, and $R \leftarrow R + \bar{A}R$ from $R = I$, the map being the class-token row of $R$; unlike attention rollout [1] it is class-specific, and unlike Chefer et al. [10] it uses no layer-wise relevance propagation; for looped transformers, loops are unrolled so each executed block contributes once. Shapley values over pieces were estimated with 16 antithetic permutations over the stimulus's own pieces; the value of a set of pieces is the log probability of the class when only those pieces are drawn, a piece being removed by drawing it in the background colour.

# C Turning the inducers

Each display keeps the polygon, inducer radius and position of a Kanizsa figure in the battery; only the inducers turn, each about its own centre. The training outlines already cover every rotation and the size of the shape the inducers specify (circumradius $4.5P$, ±15%), so that shape is familiar

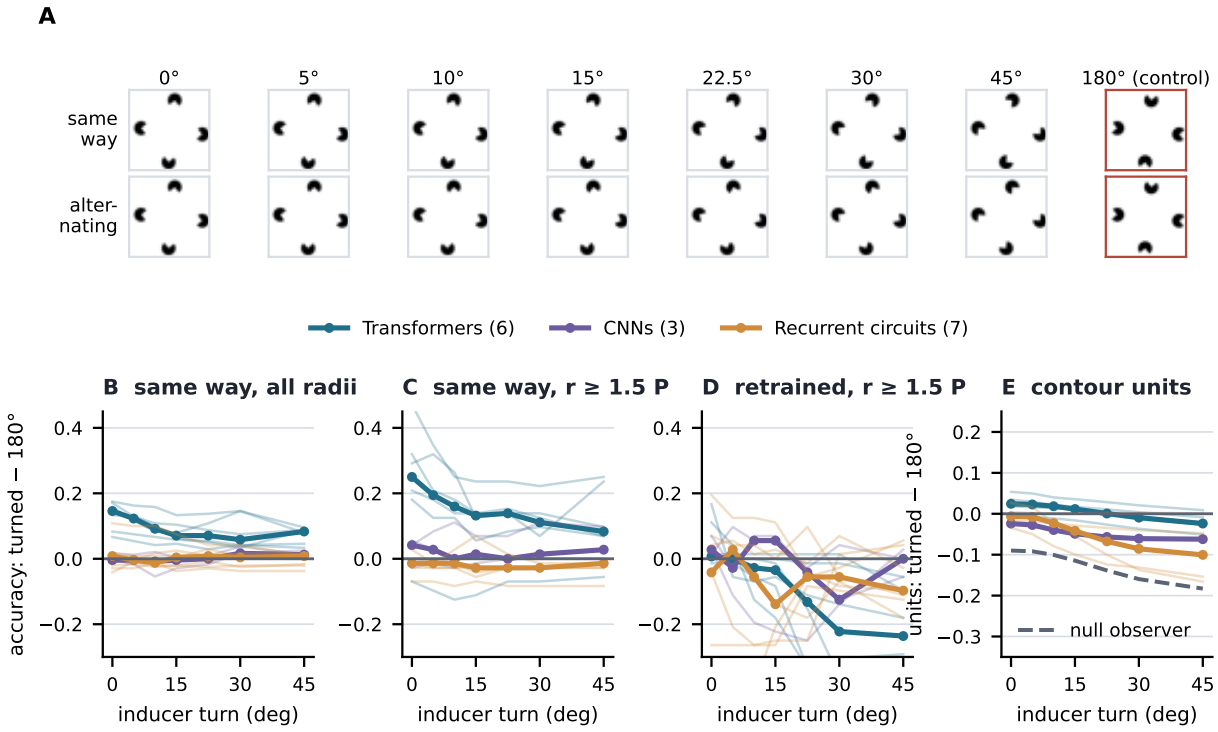



**Figure 18:** Turning the inducers. (A) Every inducer turned in place by 0–45° (0°: the Kanizsa figure), the same way (top; the fat/thin displays of Ringach and Shapley [38]) or in alternating directions (bottom), against the 180° control of the main battery (red frame); a square with inducer radius 1 patch is shown. (B–D) Accuracy with turned inducers minus accuracy with the 180° control, inducers turned the same way: all radii (B), radius $\geq 1.5$ patches (C), and the same networks retrained with inducers in their training images (D). Bold: median of each family; thin: each network. (E) The 5% most contour-selective units of each network, in the empty cells on the straight edges of the aligned figure, scaled so that an empty cell is 0 and a drawn contour 1; dashed, a null observer (oriented edge energy pooled over half a patch, scaled the same way).

to every network. In transformers the Kanizsa advantage over the 180° control falls from +0.15 (median, all radii) to +0.07 at 15° and is +0.08 at 45°; at radius $\geq 1.5$ patches, where the inducers nearly touch, it falls from +0.25 to +0.13 and +0.08 (fine-tuned ViT-B/16: +0.47, +0.24, +0.25; Fig. 18). Turned in alternating directions, the transformers' advantage is +0.02 at 15° and +0.01 at 45°. In CNNs and recurrent circuits no turn moves accuracy away from the control by more than 0.11. An advantage that survives a 45° turn in the same direction does not depend on the inducer edges lining up with one another. The contour units on the edges fall by a median 0.04 from 0° to 45° (14 networks whose units respond to faint real lines), less than the null observer (0.09). Retrained with inducers in half of the training outlines (aligned or turned 180°, at the battery's radii), the median advantage at radius $\geq 1.5$ patches is +0.01 with aligned inducers, and transformers score below the 180° control at 45° (-0.24).

## D Shapley interactions over the inducers

Shapley values of image regions cannot test completion: removing an empty region replaces background with background, so its value is zero by construction. What closure predicts instead is an interaction among the inducers: a set of aligned inducers should be worth more than its members

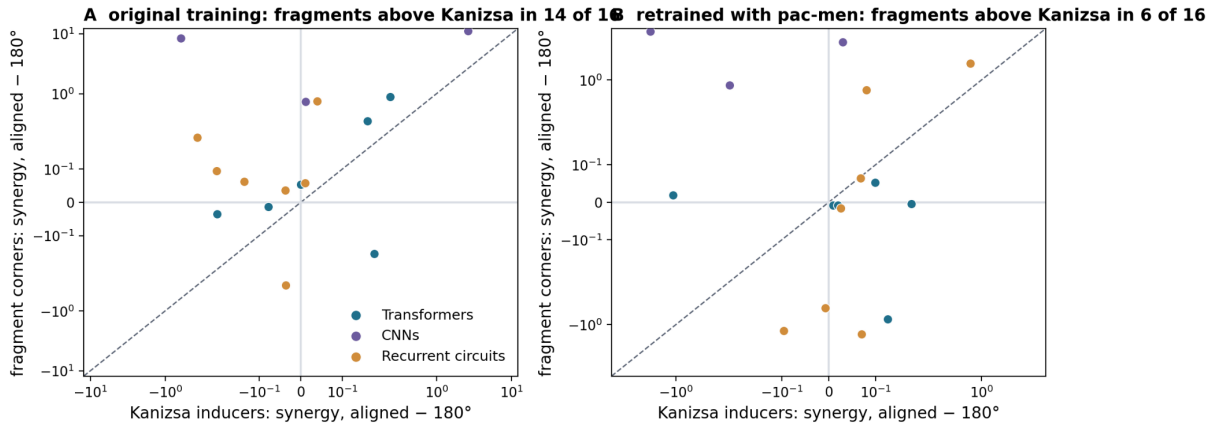



**Figure 19:** Whole-minus-parts synergy of the inducers, $v(N) - v(\emptyset) - \sum_i \big(v(\{i\}) - v(\emptyset)\big)$ with $v$ the log probability of the true class, aligned minus the 180° control (median over figures; symmetric log axes), for fragment corners (vertical) against Kanizsa inducers (horizontal), one dot per network. (A) Original training. (B) The same networks retrained with inducers in their training images.

alone. Taking the $n$ inducers of a polygon ($n = 3$–6, radii $\frac{1}{2}$–2 patches) as players, we scored every subset by the log probability of the true class (at most $2^6$ images per figure, so the values are exact) and compared aligned inducers with the same inducers turned 180° in place. As a positive control we ran the same analysis on the $n$ corner pieces of mid-gap fragments (gaps of 1 and 2 patches), aligned or each turned 180° about its own pixel centroid. In the 16 networks trained on outlines, fragment corners are superadditive relative to their control (median synergy +0.08 nats, positive in 12), Kanizsa inducers are not (-0.02, positive in 7), and fragments exceed Kanizsa figures in 14 of 16 networks (Wilcoxon $p = 0.009$; Fig. 19A). The exception is the transformers, whose aligned inducers add a little synergy (median +0.08, fragments +0.02), in line with their small Kanizsa advantage, which survives turning the inducers (Appendix C). Pairwise interaction indices [17] of neighbouring inducers, which share an edge of the shape they specify, do not separate fragments from Kanizsa figures ($p = 0.19$). After retraining with notched discs, fragments no longer exceed Kanizsa figures (6 of 16; Fig. 19B).

# E Cued-object segmentation circuits

*Architecture.* We followed the Methods and Table S1 of Mollard et al. [33]. The backbone is the Panoptic FPN with a ResNet-101 trained on COCO panoptic segmentation [24] (Detectron2 model-zoo weights), frozen; only its stride-4 map (256 channels) is used, reduced to 64 channels by a $1 \times 1$ convolution. Four horizontal hGRU levels at strides 4, 8, 16 and 32 (64, 128, 256 and 512 channels, $5 \times 5$ horizontal kernels) are linked bottom-up by blocks of two or four $1 \times 1$ convolutions with group normalization and ReLU, each followed by $2 \times 2$ max-pooling, and top-down by three hGRUs with $1 \times 1$ kernels whose drive is the upsampled higher level (bilinear upsampling, group normalization, two $1 \times 1$ convolutions with ReLU) and whose output is mixed with the old state, $H_l = (1 - \sigma(\beta_l))\,\mathrm{hGRU}(H_l, H_{l+1}) + \sigma(\beta_l) H_l$. Each hGRU computes $G_S = \sigma(U_S * H)$, $C_S = \mathrm{GN}(W_S * (H \odot G_S))$, $S = [Z - [(\alpha H + \mu) \odot C_S]_+]_+$, $G_F = \sigma(U_F * S)$, $C_F = \mathrm{GN}(W_F * S)$, $\tilde{H} = [\nu(C_F + S) + \omega(C_F \odot S)]_+$, $H' = (1 - G_F) \odot H + G_F \odot \tilde{H}$, with $U$ $1 \times 1$ convolutions, $\alpha, \mu, \nu, \omega$ learnable scalars and $[\cdot]_+$ softplus (Table S1). The network runs for 40 timesteps; a $1 \times 1$ convolution to 128 channels, ReLU and a $1 \times 1$ convolution on the lowest level read out 14 classes (background

and 13 object classes).

*Training.* Every network is trained to give every pixel of the cued object its class and every other pixel the background class, with per-pixel cross-entropy and the Lipschitz coefficient penalty of C-RBP [29] (weight $10^{-4}$, as in the paper), by C-RBP: 39 steps without gradients, one with, and the gradient at the fixed point from a Neumann series of up to 15 vector–Jacobian products that stops, as in the reference implementation, when a term grows, the sum's norm exceeds 1 or a term becomes negligible; Adam, batch 28, 21,000 iterations, learning rate warmed up to $10^{-4}$ over 1,000 iterations and multiplied by 0.8 at 10,000 and 20,000; $224 \times 224$ images, objects of at least 5,000 pixels, three random cue positions per object, the class *person* subsampled to one in six. Single-step controls are trained by ordinary back-propagation through their one step, without the penalty.

Where the paper is silent or we could not follow it: (i) the cue enters as an extra input, a disc of radius 8 px added after the $1 \times 1$ reduction; (ii) we trained on all COCO training images with objects of the 13 most frequent classes after subsampling (person, dining table, chair, car, bowl, couch, cup, truck, cat, bus, bed, motorcycle, train), not on the image set of Adeli et al. that the paper used, and evaluated on 2,000 COCO validation objects rather than on that set; (iii) group normalization uses 8 groups, and the network has about 20 million trainable parameters against about 8 million in the paper; (iv) eq. 4 of the paper has no inner rectification, $S = [Z - (\alpha H + \mu) \odot C_S]_+$; since $C_S$ is group-normalized and can be negative, the suppression then excites, and from some initialisations the state grew without bound and training reached non-finite values within 12 iterations, so we rectify the suppressive term as in the hGRU the paper cites [28]; (v) initial values (orthogonal kernels, gate biases initialised as in our fGRU, $\alpha = \omega = 0.1$, $\mu = \nu = 0$, $h_0 = 0$) are not given in the paper; (vi) the frozen backbone runs in bfloat16, the recurrence in 32-bit floating point; (vii) we clip gradients at norm 10 and use a guard that restores the last checkpoint and halves the learning rate if the loss is not finite or the penalty exceeds 20 times its recent median (in the first 20 iterations, 20 times its first value and at least 1,000), and skips a batch whose cross-entropy exceeds 25 while the penalty is normal; it fired in two runs ($9 \times 9$ kernel: iterations 18,630 and 20,756; fGRU on ViTDet: iteration 20,780) and never skipped a batch. We trained one network per setting; the paper trained five.

*Variants.* (a) Penalty weight 100, the value given in the documentation of the reference C-RBP code. (b) Finest-level horizontal kernels of $3 \times 3$, $9 \times 9$ and $15 \times 15$ (12, 36 and 60 px at stride 4) in place of $5 \times 5$; the other levels keep $5 \times 5$. In the $9 \times 9$ run the penalty rose from about 0.2 at iteration 12,000 to about 18 at 18,000, and the network reaches IoU 0.28 (0.18 on objects under 5% of the image); its checkpoint at iteration 12,000 reaches 0.25 (0.10) and labels 0.02 of the filled hexagon. (c) The fGRU cell of our other circuits (Section 3; group normalization in place of instance normalization) in place of the hGRU, with a gated top-down term added to the lower level in place of the paper's mixing, max-pooling and a $3 \times 3$ convolution between levels and a 64-channel read-out. (d) The same fGRU network on the stride-4 map (256 channels) of ViTDet ViT-B, a plain ViT pre-trained with masked autoencoding and trained with Mask R-CNN on COCO [26] (Detectron2 weights). (e) A looped transformer: the 64-channel drive at stride 4 is embedded in $2 \times 2$ patches into 784 tokens of width 512 with a learned position embedding, and one pre-norm transformer block (8 heads, MLP width 2,048) is applied 40 times to the state plus the embedded input, $h_{t+1} = \mathrm{Block}(h_t + e)$, $h_0 = 0$; a layer norm, a linear map to 64 channels, upsampling to stride 4 and the read-out of (c) follow. Attention is written out because fused attention kernels have no double backward, which the penalty needs; the network has 3.7 million parameters. (f) Single-step controls of the hGRU network and of (e), trained and read out after one step.

*Displays and measures.* For the battery test each display ($224 \times 224$, the polygon of circumradius $4.5P$ centred, $P = 16$ px, $n = 3$–6, inducer radius 1, 1.5 and 2 patches, four orientations) was shown with the cue at the polygon's centre, and the network's segmentation (argmax over classes other

than background, at the read-out step) was compared with the polygon. For the gap test, regular pentagons and hexagons of circumradius 3.75 and 4.5 patches (the main text reports the hexagon of 4.5 patches) were drawn with inducers of radius $r = s\,L/2$ at every vertex, for support ratios $s = 0.50$–$0.95$ ($L$ the side length, gaps $(1 - s)L$), four orientations. Controls keep the vertices and $r$: rotated inducers (turned 180° in place), full discs, segments 8 px thick along both sides from every vertex, of length $r$ (corners), the aligned inducers with the stretch of each side between them drawn as a 2 px line, solid or dashed (4 px dashes, 4 px spaces), a blank page and a mid-grey filled polygon. Fill is the share of the polygon's white pixels, beyond one patch from the cue, that the network labels as the cued object; the strip just outside a gap lies 0.25–1 patch outside a side, along the gap's span. Closing the gaps with 8 px bars instead of 2 px lines stops the label in every 40-step network (at most 0.44), so the thin lines serve as the closed-gap control. For reversed contrast the displays' grey levels are inverted before the backbone and the regions are taken from the original display.

*Earlier fGRU networks.* Two networks with fGRU units on Panoptic FPN features, one with the paper's settings and one with object pixels weighted three times in the cross-entropy, were trained before the corrections above: their Neumann series summed all 15 terms. In the first, the guard fired once, at iteration 19,038, so its last 2,000 iterations used half the learning rate; a run before it, without the guard, diverged at about iteration 16,000. On COCO validation objects they reach IoU 0.43 and 0.50 (0.35 and 0.35 for objects under 5% of the image, 0.50 and 0.71 above 30%; Fig. 20). In the battery's figures (48 figures) both segment the filled surface (mean IoU 0.48 and 0.55) but barely the Kanizsa figure (0.03 and 0.10), little more than with rotated inducers (0.02 and 0.07) or on a blank page (0.01 and 0.03), and over 40 timesteps the cue's effect on the lowest level, $e_t = \mathrm{mean}_c|h_t(\text{cue}) - h_t(\text{no cue})|$, spreads past the polygon, with indistinguishable maps for aligned and rotated inducers (Fig. 20A); a thin real outline is not segmented either (0.01 and 0.08), and a display combining it with aligned inducers reaches 0.10 and 0.17. Pooled over polygons with $n = 3$–$6$ sides at three sizes (circumradius 3–4.5 patches), neither network fills more than 0.18 and 0.10 of any Kanizsa figure at support ratios up to 0.70; fills of 30% or more begin at 0.75 and 0.80 (Fig. 21B,C), and at support ratios of 0.80 and above full discs fill as much as aligned inducers on average (paper settings 0.47 against 0.34; weighted loss 0.36 against 0.37). These pooled means hide the dependence on size: in the large hexagon the network trained with the weighted loss fills 0.68 of the Kanizsa hexagon with 7 px gaps and 0.19 between full discs (Table 1). Across the five notched inducers of the configural-effect shape audit (Fig. 15B), which specify the same polygon, aligned notched crosses fill nothing (0.00 and 0.00), aligned rings fill less than rings without their notch (0.26 against 0.68; 0.09 against 0.30), and only notched squares and hexagons fill more aligned than without their notch (squares 0.18 against 0.02 and 0.26 against 0.10; support ratio $\geq 0.8$, displays whose overlapping inducers seal the interior, 12%, excluded; Fig. 21D,E).

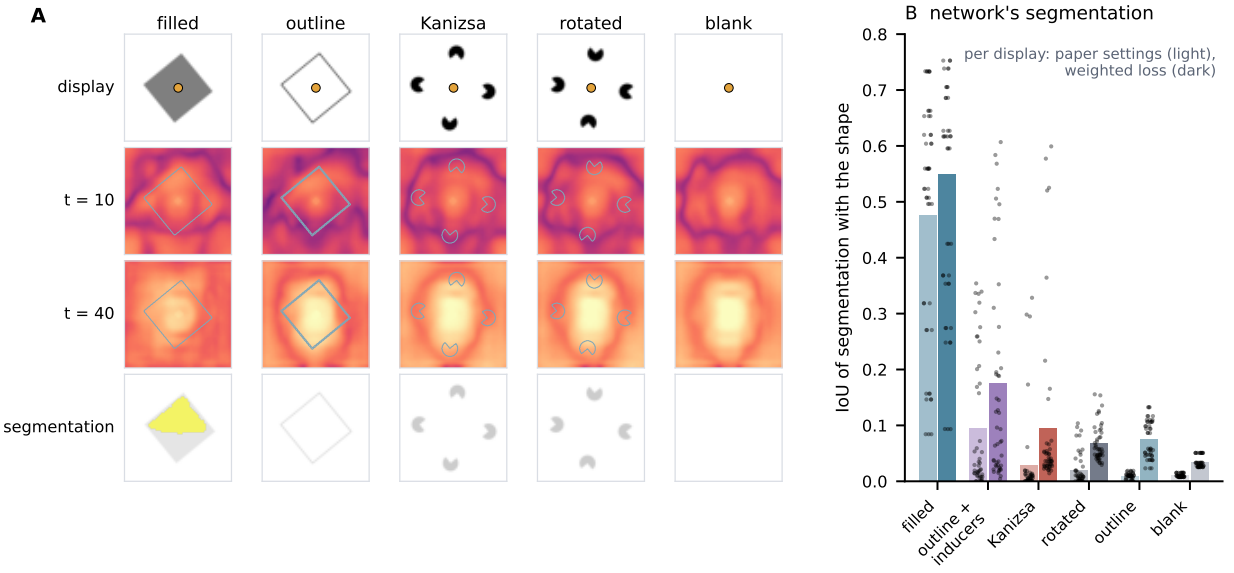



**Figure 20:** The earlier fGRU networks on the battery, cue (orange) at the centre of a polygon. (A) Paper settings, square with inducer radius 1 patch: displays; the cue's effect on the lowest recurrent level, $|h_t(\text{cue}) - h_t(\text{no cue})|$ averaged over channels, after 10 and 40 timesteps (one logarithmic colour scale for all maps; blue lines, the display's contour pixels); and the region the network labels as the cued object (yellow). (B) Intersection over union of that region with the polygon for each display ($n = 3$–$6$ sides, inducer radius 1–2 patches, four orientations); bars, mean for the network trained with the paper's settings (light) and with object pixels weighted three times in the loss (dark); dots, single figures.

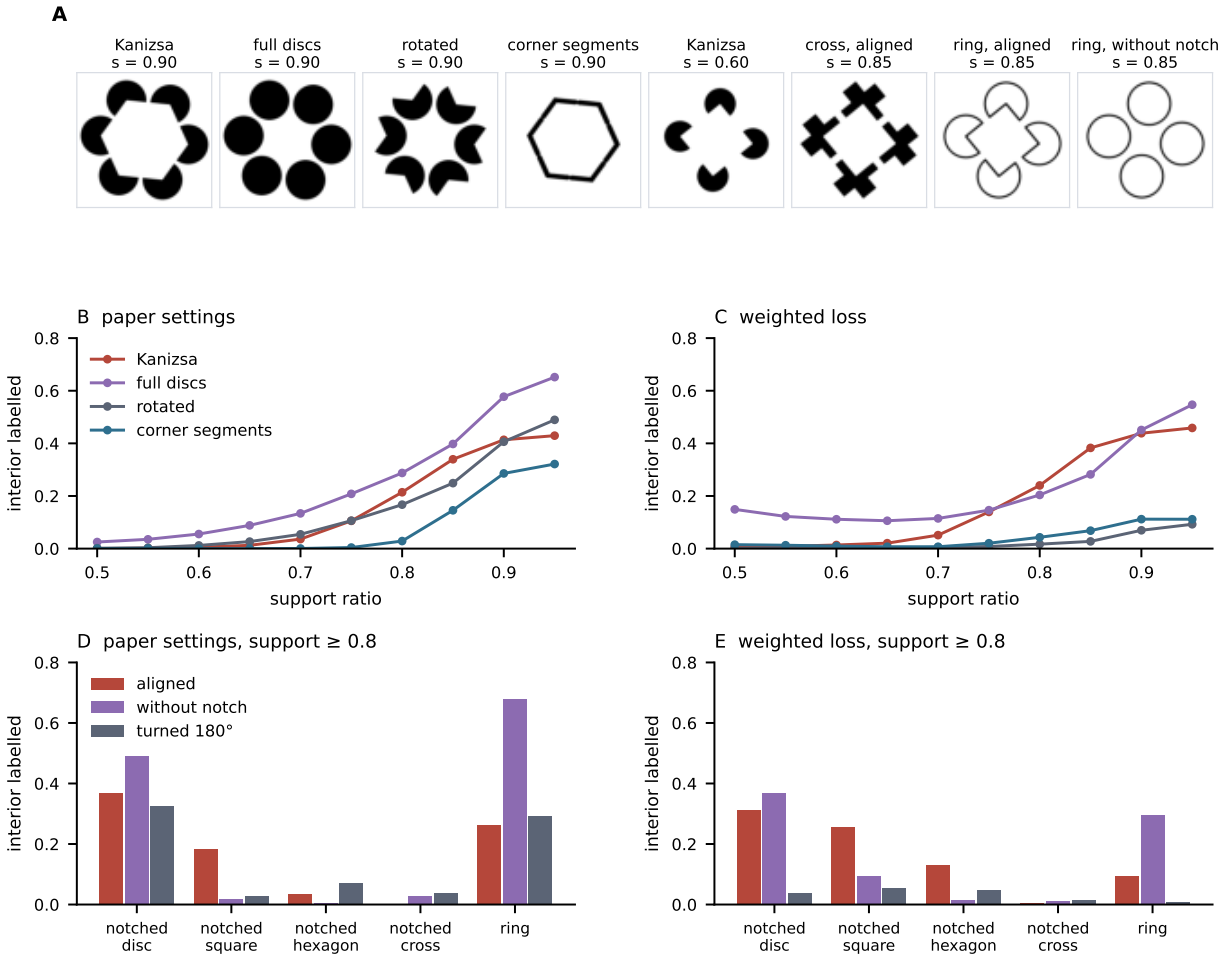



**Figure 21:** Support ratio and inducer shape in the earlier fGRU networks. (A) Examples: a hexagon at support ratio $s = 0.90$ shown with aligned inducers, full discs, rotated inducers and thick corner segments; a Kanizsa square at $s = 0.60$; aligned notched crosses, aligned rings and rings without their notch at $s = 0.85$. (B, C) Share of the polygon's white pixels labelled as the cued object (fill), against support ratio ($n = 3$–$6$, circumradius 3–4.5 patches, four orientations), for the network trained with the paper's settings (B) and with the weighted loss (C). (D, E) Fill at $s \geq 0.8$ for five notched inducers that specify the same polygons: aligned, without the notch and turned 180° about the vertex; displays whose overlapping inducers seal the interior are excluded.